

Trading volume manipulation and competition among centralized crypto exchanges

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Abstract

How competition affects manipulation by firms of information about important attributes of their products and how such information manipulation impacts firms' short-term and long-term performance are open empirical questions. We use a setting that is especially suitable for answering these questions—centralized crypto exchanges, on which information manipulation takes the form of inflated trading volume. We find that static and dynamic competition measures are positively associated with volume inflation, indicating that competition may lead to increased information manipulation. Exchanges that manipulate volume obtain short-run benefits but are punished in the long run, consistent with the tradeoff between short-lived increases in rents and future losses due to damaged reputation.

Keywords: Crypto exchanges, volume inflation, competition, information manipulation, regulation.

JEL classification: G18, G23, L10, L13, L15.

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1 Introduction

Information manipulation by firms with the goal of misrepresenting their products' attributes is prevalent in many settings.¹ Information manipulation may be a rational strategy in a setting characterized by asymmetric information (e.g., [Frankel and Kartik \(2019\)](#), [Ball \(2022\)](#), and [Sun \(2022\)](#)).

In this paper, we ask two questions. The first is whether information manipulation by firms is related to competition among them. Theoretically, the competitive environment in which firms operate has been shown to impact their incentives to manipulate information (e.g., [Mayzlin et al. \(2014\)](#) and [Grunewald and Krakel \(2022\)](#)). Information manipulation incentives are expected to be higher for firms operating under more intense competition. Empirically, no consensus with regards to this question has been reached so far. For example, while [Luca and Zervas \(2016\)](#) and [Becker and Milbourn \(2011\)](#) find that competition is positively associated with information manipulation in the cases of restaurants and credit rating agencies, respectively, [Zinman and Zitzewitz \(2016\)](#) do not find robust relation between the two. In our setting, in which competition can be measured very granularly, we find that firms indeed tend to manipulate information to a larger degree when facing more intense competition.

Our second question, the answer to which has so far been elusive, is whether the benefits of information manipulation are higher than its reputational costs. We hypothesize that a firm's decision to what extent to attempt to mislead its customers about its products' attributes involves a tradeoff between higher short-term gains and lower future profits due to damaged reputation. [Karpoff, Lee, and Martin \(2008\)](#) show that the market punishes firms that were found to be misreporting financial information. We find that information manipulation tends to be beneficial to manipulating firms in the short run and harmful in the longer run, suggesting that the market can gradually discover misrepresentation of information and that information manipulation has real, likely reputational, effects, which offset its shorter-lived gains.

¹See e.g., [Luca and Zervas \(2016\)](#) for the case of fake reviews of restaurants, [Mayzlin, Dover, and Chevalier \(2014\)](#) for the case of fake reviews of hotels, [Zinman and Zitzewitz \(2016\)](#) for the case of misreporting the amount of snow by ski resorts, and [Becker and Milbourn \(2011\)](#) for the case of credit rating inflation.

We focus on a particular industry—centralized crypto exchanges, whose role is to facilitate trades in crypto assets. This industry is especially suitable for examining determinants and consequences of information manipulation in a competitive setting for several reasons. First, exchanges have access to an easily implementable technology for manipulating information regarding one of their most important attributes: trading volume. The importance of trading volume to potential traders is profound, as higher trading volume implies a deeper order book and lower expected trade execution costs.

Trading volume on crypto exchanges can be artificially inflated in two ways. First, an exchange can execute back-and-forth transactions among its own accounts by making an order and immediately taking it, usually with help of third parties.² Such orders appear in an order book for a split second but cannot be taken by independent, non-exchange-affiliated traders. Second, an exchange can create perverse incentives for traders and/or project owners to trade with themselves, e.g., by eliminating trading commissions for “top-tier” traders or by requiring projects to facilitate a certain minimal trading volume to maintain exchange listing, which the projects typically do with the help of market makers.³ In neither of the two cases can fake trading increase order book depth or improve trade execution for an unaffiliated liquidity trader. What fake trading does is: 1) it raises the realized price impact relative to an anticipated one, and 2) it increases uncertainty regarding price impact. Our examination of fake trading on crypto exchanges builds on and expands the pioneering work of Cong, Li, Tang, and Yang (2022).

The importance of trading volume as an attribute of a crypto exchange’s product offering is even higher in the presence of data aggregators, such as [www.CoinMarketCap.com](https://www.coinmarketcap.com) and [www.CoinGecko.com](https://www.coingecko.com), which explicitly rank exchanges based, in part, on trading volume. In the absence of credible estimates of expected trading costs on exchanges, many traders rely on these rankings in their choice

²See SEC’s wash trading allegations of Binance as the most recent example of possible fake trading facilitated by crypto exchanges: <https://www.bloomberg.com/news/articles/2023-06-05/latest-binance-charges-inside-the-sec-s-wash-trading-allegations#xj4y7vzkg>.

³Requirements for minimal trading volume frequently appear in official listing proposals presented by exchanges to projects considering listing.

of a trading venue. Importantly, while wash trading is forbidden in most asset markets,⁴ the legality of wash trading on crypto exchanges is a gray area, due to limited regulation of crypto markets in most jurisdictions and to the cross-border nature of most crypto exchanges. Importantly, in our setting, it is the crypto exchanges themselves that may have the incentives to inflate volume. In contrast, in traditional markets, it is usually investors who wash trades, e.g., for tax purposes (e.g., [Grinblatt and Keloharju \(2004\)](#)).

An additional reason for the suitability of the crypto exchange industry for studying firms' information manipulation choices is that there are two crucial elements underlying the relation between information manipulation choice and reputation: the value of an important attribute of a firm's product has to be unobservable ex-ante, and it has to be decipherable ex-post, at least partially. Both these elements are present in the crypto exchange setting: the true depth of order books and trading costs (e.g., price impact) can only be estimated gradually over time.

Moreover, the crypto exchange industry enables a more nuanced analysis of the relation between competition and information manipulation than most other settings. Although young, the industry is very dynamic, with numerous entries and several exits since the establishment of the first crypto exchanges in 2010.⁵ Competition among crypto exchanges has several layers. On the one hand, exchanges are heterogeneous both operationally—in terms of the set of currency pairs traded on them, and geographically—catering to partially overlapping sets of clients. On the other hand, competition among multiple trading platforms within a given currency pair is competition in largely homogeneous goods.

Finally, the tradeoff between short-term gains to information manipulation and longer-term reputational costs is especially pronounced in the setting of centralized crypto exchanges due to high

⁴For example, in the United States, wash trading was made illegal after the passage of U.S. Securities and Exchange Act in 1934 and the Commodity Exchange Act in 1936.

⁵Bitcoinmarket.com went live in March 2010, followed in July 2010 by Mt. Gox. Both exchanges are now defunct. According to www.CoinMarketCap.com, there are 236 crypto exchanges specializing in spot markets as of February 5, 2024.

volatility, lack of regulation, and other risks in this market (e.g., [Harvey and Rabetti \(2023\)](#), [Landsman, Lyandres, and Rabetti \(2023\)](#), and [Cong, Prasad, and Rabetti \(2024\)](#))).

Trading volume manipulation is unobservable and not directly detectable without information on addresses of accounts that performed each trade—the type of information that is known only to exchanges themselves. To estimate the extent of volume inflation, we proceed in two steps. First, we estimate statistical measures of anomalous trading volume at the most granular level possible, i.e., at the exchange-pair-month level. Second, we examine whether our estimates of anomalous trading volume are associated with exchanges’ incentives to fake trades. A positive relation between anomalous volume and such incentives would plausibly suggest that anomalous volume is at least partially due to volume inflation by exchanges.

Our first measure, also used in [Amiram, Bozanic, and Rouen \(2015\)](#) in the context of detection of errors in financial statements, in [Michalski and Stoltz \(2013\)](#) in the context of detection of errors in macroeconomic data, and, most relevant for our study, in [Cong et al. \(2022\)](#) for detection of wash trading on crypto exchanges, is deviations of the frequencies of first digits of trade sizes from Benford’s Law ([Benford \(1938\)](#))).

Our second measure, which is novel to the finance literature to the best of our knowledge, is based on a machine-learning non-parametric algorithm called EDM (E-Divisive with medians), developed by Twitter, which identifies structural breaks in data series. The larger the number of identified structural breaks in the trading volume or number of trades series, the higher the chance of data manipulation.

Since these statistical measures aim to identify particular deviations from normal trading patterns and neither of them is likely to be capable of identifying the majority of such deviations, we aggregate these measures by computing their principal components. We use both the principal-component-based measure and the individual measures in the empirical tests.

We perform several analyses with the goal of linking estimated anomalous volume to unobservable fake trading. First, consistent with findings in other industries that firms with low reputation are

more likely to manipulate information (e.g., [Luca and Zervas \(2016\)](#)), we show that our estimates of anomalous volume tend to be the lowest for exchanges with the highest web popularity, for the oldest exchanges, and for exchanges that are regulated to some degree, i.e. for exchanges that are expected to have the lowest incentives and/or ability to manipulate trading volume. Second, we find that anomalous volume measures are higher for exchanges with relatively low levels of self-imposed regulation and compliance, consistent with anomalous volume estimates being related to the ease of manipulation.

Third, we use a pseudo-natural experiment—the ban on crypto exchanges in China in September 2017, which led to their exodus to other locations—mostly to Hong Kong and the rest of Pacific Asia. We employ a difference-in-differences methodology to compare changes in anomalous volume measures for exchanges that were subject to the ban in China to those that were not subject to the ban. We consider the ban as an exogenous shock that affected the ability of Chinese exchanges to inflate trading volume. In particular, we posit that relocating exchanges overseas presents operational challenges that could temporarily limit exchanges’ resources available for volume inflation. We find that exchanges that moved out of China following the ban experienced a several-months-long reduction in anomalous volume estimates relative to control samples.

Fourth, we use another exogenous shock—the introduction by CoinMarketCap, the largest crypto data aggregator—of a new exchange quality index in December 2019, which reduced the benefits of fake trading by explicitly penalizing exchanges with large estimated volume inflation. We investigate changes in anomalous volume estimates following the introduction of the CoinMarketCap exchange quality index. Before introducing the index, CoinMarketCap ranked exchanges based on their reported volume. After introducing the index, CoinMarketCap replaced reported volume with a proprietary liquidity score that is less likely to be affected by fake volume. We hypothesize that introducing an index that reduces the benefits of volume inflation by mitigating its impact on exchange rankings will likely result in a decrease in fake trading by exchanges. If our anomalous volume estimates capture fake trading, we expect the decrease in anomalous volume to be more pronounced for

a) exchanges whose ranking was impacted in the most negative way following CoinMarketCap index introduction and b) exchanges with the largest estimated anomalous volume prior to the index introduction. Consequently, we define treated exchanges as a) those with the largest drop in rankings in the three months following the index introduction relative to three previous months and b) those with top-five anomalous volume estimates in the three months before December 2019. Control exchanges are those with a) largest gains in rankings and b) bottom-five pre-index-introduction anomalous volume estimates. The results of the analyses of the Chinese ban and CoinMarketCap index introduction suggest that our anomalous volume measures are indeed correlated with exchanges' incentives/ability to inflate trading volume, and as a result, are likely to capture, at least to a certain extent, unobservable volume inflation.

Having related our estimated measures of anomalous volume to unobservable volume inflation, we use these measures to examine the relation between static and dynamic competition among exchanges on the one hand and the extent of fake trading on them on the other hand. Our most important finding with respect to static competition is that exchanges do not fake trading equally in all currency pairs: Pairs especially prone to volume manipulation are those that are being traded on many exchanges, i.e. pairs in which competition among exchanges is strong.

We then examine how dynamic competition environment, i.e., changes in exchanges' competitive landscape, influences fake trading on them. In particular, we analyze how an entry of an exchange into a market of a particular currency pair impacts the extent of fake trading on incumbent exchanges, i.e. exchanges on which this pair was traded previously. We find that entry by a competitor tends to increase incumbents' fake trading estimates at the exchange-pair level. This evidence is in line with the predicted relation between competition and incentives for unethical behavior (e.g., [Shleifer \(2004\)](#)), and the results of [Becker and Milbourn \(2011\)](#) for the case of inflation of firms' credit ratings and [Luca and Zervas \(2016\)](#) for the case of manipulation of restaurant reviews.

The last part of our analysis concerns short-run and longer-run operational effects of fake trading. We examine the effectiveness of volume inflation in light of the conjectured tradeoff between short-term

gains versus long-term losses due to harmed reputation. We measure real outcomes in two ways. The first is the exchange’s web popularity, i.e. the number of visits to an exchange’s website. The second is the exchange’s social media popularity, as measured by retweets of Twitter tweets by the exchange’s account. Both outcomes are likely correlated with (legitimate) trading on an exchange and with resulting revenues of the exchange. The overall finding is that volume inflation has a positive impact on web and social media popularity of an exchange and its share of trades in the short run (three to six months) and a negative impact in the longer run (12 to 24 months).

Our paper makes two main sets of contributions. Although we use a unique setting, our first set of contributions is quite general. In addition to providing an out-of-sample corroboration of the result in [Becker and Milbourn \(2011\)](#) and [Luca and Zervas \(2016\)](#) that increased competition leads to higher information manipulation, we contribute to the literature on information manipulation in the presence of reputational concerns by examining the effectiveness of information manipulation, i.e. we analyze whether fake trading by crypto exchanges has real, operational benefits. In addition, we analyze whether there is a tradeoff between short-term benefits and long-term reputational costs to the manipulation of information. Our finding that exchanges that inflate trading volume perform well in the short run but are punished in the long run is consistent with the model of [Mailath and Samuelson \(2001\)](#), in which reputation is built gradually and dissipates gradually.

Our second contribution is to the nascent literature examining the behavior of crypto exchanges, which are one of the more important players in the crypto market (e.g., [Amiram, Jørgensen, and Rabetti \(2022\)](#), [Cong et al. \(2022\)](#), [Cong, Landsman, Maydew, and Rabetti \(2023\)](#), and [Griffin and Shams \(2020\)](#)). We complement [Cong et al. \(2022\)](#), who perform the first large-scale analysis of fake trading on crypto exchanges, along the following dimensions. First, we corroborate their findings that a significant fraction of trading in four major cryptocurrencies against USD. is likely fake. We show that these results continue to hold in a sample that includes almost the entire population of most important cryptocurrencies. Second, we analyze market-level, exchange-level, and currency pair-level characteristics that are associated with the extent of fake trading, focusing on competition-

related characteristics. Third, we examine the implications of fake trading on short-term and longer-term measures of exchanges’ performance. Finally, we introduce a novel, machine-learning-based statistical measure of fake trading. Our findings specific to crypto exchanges provide timely insights for regulation of crypto markets in light of growing evidence of the prevalence of financial misconduct and cyber crime in these markets.⁶

The remainder of the paper is organized as follows. The next section describes our data sources. In Section 3, we characterize the main empirical regularities of the crypto exchange industry. In Section 4, we describe our measures of anomalous trading. In Section 5, we perform several analyses linking our anomalous trading measures to unobservable volume inflation. In Section 6, we examine the relation between static and dynamic competition among exchanges on one hand and fake trading on them on the other hand. Section 7 examines short-term and longer-term real effects of fake trading. Section 8 concludes. Appendix A contains illustrations of various anomalous trading patterns and a detailed description of methods for anomalous trading detection. Appendix B contains variable definitions and robustness tests.

2 Data

2.1 Main data source

Our main data source is Kaiko database (www.Kaiko.com), which provides details of every executed transaction on over 100 crypto exchanges, including currency pair, price, volume, and timestamp.⁷

The choice of exchange coverage by Kaiko incorporates the following aspects:

Compliance: Soundness of exchange policies, such as KYC/AML, market surveillance, country risk, financial regulation, and insurance coverage.

⁶See e.g., [Cong et al. \(2023\)](#) for evidence of tax avoidance and [Cong, Harvey, Rabetti, and Wu \(2023\)](#) for evidence of crypto-enabled cyber crimes.

⁷Kaiko collects, normalizes, and stores terabytes of historical cryptocurrency data from dozens of spot and derivatives exchanges. The procedure consists of retrieving the data using an exchange’s API or taking frequent snapshots of its web-based platform.

Business: Maturity and quality of the exchange’s business, measured by the exchange’s age, the strength of the team, and the product offerings.

Technology: The capacity of the exchange’s platform to provide market-level information through widely adopted distribution standards, while ensuring sufficient level of infrastructure reliability.

Data Quality: The breadth of public market data (trades, OHLCV (open, high, low, close, volume), and order books), and the capacity of the exchange to provide real-time feeds and/or historical data.

Security: The capacity of the exchange to provide sufficient level of security and protection to its users while mitigating the occurrence of security breaches.

Liquidity: The level and stability of trading activity but also the quality and reliability of the liquidity publicly reported by the exchange.

We apply the following filters to form the sample of exchanges that we use in our analysis: (1) Centralized exchanges, (2) Spot trading, and (3) Must contain at least one pair at any point in time during our sample period in which the base asset is one of the following: BTC or ETH or USDT.⁸ There are 50 exchanges that satisfy these criteria. We limit the analysis to pairs that had at least 10 daily trades on average throughout our sample period. Our sample begins in May 2013, ends in May 2021 (97 months in total), and contains 237,205 exchange-pair-month observations.

2.2 Other data

We use several additional data sources. We obtain exchange-level characteristics, such as the adoption of anti-money-laundering measures (AML), the existence of know-your-customer requirement (KYC), and exchange location from www.CoinGecko.com and www.Cointelligence.com. Time

⁸These base currencies have important functions in the cryptocurrency markets. Bitcoin (BTC) is the oldest and largest cryptocurrency. Ether (ETH) is the native currency of the main smart-contract-enabling blockchain, Ethereum. Tether (USDT) is the stable coin with the highest market cap. To ensure that all relevant currency pairs are included in the analysis, we correct for inconsistencies in pair names. One type of inconsistency is pair ordering. For instance, in some exchanges the pair USDT-BTC is named as BTC-USDT. Another type of inconsistency is variations in the cryptocurrency ticker symbol. For example, Bitcoin Cash appears as BCH, BTCH or BTCASH on various exchanges.

series of web traffic to exchanges’ websites are obtained from www.Similarweb.com. Time series data on the technological development of exchanges, proxied by their code revisions on the most popular open-source platform, Github, are from www.Github.com, and time-series evolution of exchanges’ social media activity on one of the most popular social media platforms, X (formerly Twitter), is from www.Twitter.com.

3 The crypto exchange industry

Panel A of Table 1 presents summary statistics of the cryptocurrency exchange market.

[Insert Table 1 here]

The mean number of distinct cryptocurrencies traded on all exchanges (collectively) in our sample in a given month is 944, whereas at the end of our sample period, 3,112 currencies were traded. The mean number of new cryptocurrencies appearing on exchanges on a monthly basis is 85 and the mean number of currencies that exit the market in a given month is 48; the medians are much lower, 32 and 2, respectively. The mean number of distinct cryptocurrency pairs traded on exchanges in a given month is 1,390 and at the end of our sample period trading occurred in almost 4,400 distinct pairs. On average, 146 (65) currency pairs are listed (delisted) across exchanges in a given month; the medians, again, are much lower, at 62 and 6, respectively.

The mean reported aggregate monthly trading volume is \$U.S. 340 billion. There are over 400 million reported trades on average per month. The distributions of trading volume and of the number of trades are very skewed—the medians are one-to-two orders of magnitude lower than the means. Importantly, reported values of trading volume and number of trades may be highly inflated due to abundance of fake (wash) trades on crypto exchanges, documented in [Cong et al. \(2022\)](#)—a finding that we corroborate and extend in this paper.

The crypto exchange market (or, more precisely, its subset covered by Kaiko data and satisfying our selection criteria) grew from a single exchange—Bitfinex—in 2013 to 44 exchanges in the first half

of 2019. The market is quite dynamic: on average, a new exchange is established every two months. The last row in Panel A of Table 1 illustrates the competitive landscape of the crypto exchange market described by Herfindahl index, computed using the number of currencies traded on each exchange.⁹ The mean Herfindahl index is quite high (0.63); the reason is that very few exchanges operated during a large part of our sample period. To gain a better understanding of the evolution of concentration of the crypto exchange market, in Figure 1 we depict the evolution of Herfindahl index over time.

[Insert Figure 1 here]

As evident from Figure 1, the Herfindahl index has been steadily declining over time—from close to one up to 2016 (and before) to around 10% in the end of our sample period.

Panel B of Table 1 reports summary statistics at the level of a crypto exchange. A typical exchange in our sample is about two years old. The mean (median) number of currency pairs listed on an exchange in a given month is 137 (33). Every month, an exchange lists 9 new pairs on average and delists 5. The next few rows provide statistics of various characteristics of crypto exchanges that illustrate their operating features. 51% of exchange-month observations belong to exchanges that have implemented anti-money-laundering measures (AML).¹⁰ 59% of observations correspond to exchanges that have implemented know-your-customer (KYC) procedures.¹¹ There is wide variation in popularity of crypto exchanges, measured by web traffic to them, with the mean (median) of 290M (40M) visits per month. There is also significant variation in exchanges' social media activity (measured by retweets of Twitter tweets from the exchange's account) and in its technological development (measured by Github commits).¹²

⁹Note that while it is, in principle, possible to compute the Herfindahl index using other metrics, such as trading volume or number of trades, these measures are likely to be biased due to fake trading that we document. Thus, we prefer to base the measure of market concentration on a variable that is hard to manipulate.

¹⁰AML indicator equals one if the exchange has implemented an AML policy and provides detailed information about conformity with accepted international AML procedures. We use trust score from www.CoinGecko.com for all exchange characteristics discussed in this paragraph.

¹¹KYC indicator equals one if there is evidence that the exchange provides clear guidelines as to the documents required for verification of the sources of clients' funds.

¹²Deng, Lee, and Zhong (2018) develop alternative, more precise measures of technological development. Unfortunately, while being highly correlated with a commit-based measure, these measures have many missing observations in our setting, leading to low statistical power of tests (see also Lyandres, Palazzo, and Rabetti (2022)).

Panel C of Table 1 presents statistics at the level of a currency pair. Currency pairs tend to be traded on multiple exchanges. The mean number of exchanges on which a pair is traded is 4.18, and the median is 2; BTC-ETH pair was traded on 37 exchanges at its peak. The mean age of listing of a currency pair (on any exchange) is 24 months, and the mean age of listing on a particular exchange is 15 months. In 53% of cases, the quote currency is a token, whereas the rest are coins.¹³

4 Measurement of anomalous volume

Measuring fake trading is difficult, and no direct measures exist. According to Gerald Chee, head of research at [www.CoinMarketCap.com](https://www.coinmarketcap.com), “there is no way to tell if an exchange is inflating volume or not by merely looking at the volume they report. The only way to detect ‘wash trades’ would require access to ‘account-ID’ data... [and] only exchanges have access to these [data]”.

As a result, we develop indirect measures of fake trading. First, in this section, we measure anomalous (not necessarily fake) trading using statistical methods. Second, in the next section, we examine whether the extent of anomalous trading is related to exchange characteristics that are likely to be related to exchanges’ incentives and/or ability to fake trades. A presence of such relation would be consistent with a conjecture that *estimated anomalous* trading proxies for *unobservable fake* trading.

Figure A.1 in the Appendix presents some examples of anomalous trading. Several patterns of anomalous trading emerge: seeming structural breaks in trading volume and the number of trades; unusual spikes of trading activity; constant trading volume for prolonged time periods. In the next two subsections, we describe our anomalous trading measures.

¹³While both tokens and coins are used to define a unit of value, there are important differences between them. First, tokens are built on top of existing blockchains, such as Ethereum, while coins, such as Bitcoin, Ether, or Litecoin, are unique digital currencies, which are based on their own, standalone blockchains. Second, and perhaps more importantly, coins are used mostly as a method of payment or a store of value, whereas tokens are also used to activate features on decentralized applications (DApps) they were designed for. In separating crypto coins and tokens, we employ data from Lyandres et al. (2022).

4.1 Benford’s law

Our first measure of irregular trading patterns applies Benford’s Law ([Benford \(1938\)](#)), based on the likelihood of occurrences of first digits of a series.¹⁴ [Aloosh and Li \(2022\)](#) and [Gandal, Hamrick, Moore, and Oberman \(2018\)](#) use an internal book of individual trader level records leaked from Mt. Gox exchange to show that deviations from Benford’s Law are useful in detecting anomalous trading. We follow [Cong et al. \(2022\)](#) in using Benford’s Law to detect anomalous trading on crypto exchanges. To quantify deviations from Benford’s Law, we calculate mean absolute deviation (MAD) between theoretical, Benford’s Law-based fractions of leading digits and their proportions in the observed series, as described in detail in Appendix A and illustrated in Figure A.2.¹⁵

4.2 E-divisive with medians (EDM)

The second method that we use for measuring anomalous trading is rooted in the computer science literature that focuses on detection of structural breaks, i.e. mean shifts or trend shifts in a series. One particular algorithm that is suitable for detecting structural breaks in trading volume and the number of trades series is called EDM (E-Divisive with medians (e.g., [James, Kejariwal, and Matteson \(2016\)](#))). The algorithm attempts to determine whether a new chunk of time series data is considerably different from the previous chunk through comparisons of various permutations of the data. One important reason for the suitability of EDM in our setting is that it is robust to the presence of short-lived abnormalities in the series. This is particularly relevant because trading volume data tend to have occurrences of peaks, which, despite often being a result of a natural data generating process, may appear as anomalies when data series are aggregated into short intervals. In addition, EDM is non-parametric. This is important because trading volume that is partially a result of wash trading does not usually conform to a particular distribution. See Appendix A for a precise description of estimation

¹⁴In many naturally observed series, which are not constrained to a certain range, 1 appears as the leading digit in 30.1% of cases. As the value of the first digit increases, its frequency decreases. 2 (3, 4, 5, 6, 7, 8 and 9) appear as the leading digit in 17.6% (12.5%, 9.7%, 7.9%, 6.7%, 5.8%, 5.1% and 4.6%) of the cases.

¹⁵Another way to quantify the distance between theoretical and observed fractions of leading digits is using Chi-square differences, as in [Cong et al. \(2022\)](#). All our results are robust to using this measure of distance.

of EDM measure and Figure A.3 for its illustration.

4.3 Aggregating anomalous trading measures

Neither of the anomalous trading measures described above is perfect. Each measure is constructed with the goal of identifying particular deviations from normal trading, frequently observed in the data. To construct a unified measure of anomalous trading, we extract principal components (PCA) of the measures described above at the exchange-month-pair level. Utilizing PCA in our analysis has at least three key advantages over employing raw measures directly. First, PCA consolidates the majority of the original variance, proving beneficial in numerous situations in which raw measures may disagree. Second, PCA effectively addresses the issue of multicollinearity among raw measures, which is particularly significant given that we derive individual measures from volume and number of trades series that are highly correlated. Third, PCA standardizes the data, easing the interpretation of the results. Despite relying on PCA as our primary indicator of anomalous trading, we present the measures for MAD and EDM in all tables to ensure robustness. See Appendix A for a discussion of applicability of a principal-components-based measure in our setting.

Table 2 reports distributions of anomalous trading measures (PCA, MAD, and EDM), based on volume of trading, number of trades, and both volume and number of trades in the case of PCA. To facilitate the discussion of economic significance, all anomalous volume measures are normalized to have zero mean and unit variance, as in Panels B and C of Table 2.

[Insert Table 2 here]

Figure 2 presents the evolution of mean principal-component-based anomalous trading measures over time. In particular, every month, we compute an equally-weighted average over all exchange-currency pair-months of PCA-based measure. The vertical lines centered around point estimates depict confidence intervals of these estimates.

[Insert Figure 2 here]

Mean anomalous trading measures have generally been increasing since 2017. The temporal increase in anomalous trading—an interesting finding in its own right—is potentially consistent with the entry of less reputable exchanges, with volume inflation being more prevalent in newer, less liquid currency pairs, and with the temporal increase in the intensity of competition among exchanges. We examine these conjectures below.

5 Is anomalous trading a sign of fake trading?

So far we have constructed and estimated statistical measures of anomalous trading, which may or may not be a result of fake trading on/by exchanges. The next step is to examine whether these measures of anomalous trading are related to volume inflation by means of fake trading. In the next four subsections, we analyze the link between estimated anomalous trading and exchange’s incentives/ability to engage in fake trading.

5.1 Mean exchange-level anomalous trading

We start with an intuitive “smell test”, i.e. a visual examination of exchanges with high and low estimates of anomalous volume. Figure 3 presents average (over time) PCA-based measure of anomalous trading for all fifty exchanges in our sample.¹⁶

[Insert Figure 3 here]

Several findings emerge from Figure 3. Exchanges widely regarded as most reputable—Binance and Coinbase—have below-sample-mean measures of anomalous trading. In addition, three exchanges in our sample that are regulated by the New York State Department of Financial Services—Coinbase, Bitflyer, and Gemini—have some of the lowest anomalous trading measures. On the other hand, some of the exchanges with the highest estimated anomalous trading measures—Bibox, CoinEx, Huobi, and

¹⁶Illustrations of anomalous trading estimates using MAD-based and EDM-based measure are qualitatively similar.

OKX—have been subject to negative press coverage that may be indicative of subpar governance.¹⁷

5.2 Anomalous trading, exchange compliance, and activity

In Table 3, we examine the relations between measures of anomalous trading volume and various exchange characteristics.¹⁸ The focus is on characteristics related to exchanges’ ability/incentives to inflate trading volume: measures of exchanges’ (exogenous or self-imposed) compliance and the amount of real activity on exchanges. In this analysis, we also control for additional exchange characteristics, such as exchange’s age and the number of currency pairs traded on it. Since the relations are estimated at the exchange-pair-month level, we also control for pair-level characteristics.

[Insert Table 3 here]

Exchanges with anti-money-laundering (AML) provisions in place have significantly lower mean anomalous trading measures than those without such provisions—the difference in means between the two subsets of exchanges corresponds to 0.25-0.83 standard deviations of anomalous volume measures. Similarly, exchanges with know-your-customer (KYC) requirements have significantly lower anomalous trading measures than exchanges without such requirements—the difference corresponds to 0.10-0.41 standard deviations of anomalous volume measures. Moving to the relation between anomalous trading volume on an exchange and the real activity on it, we observe that web traffic, social media popularity, and technological activity are all negatively related to anomalous volume, i.e. struggling exchanges are more likely to exhibit anomalous trading volume. Overall, the relations between measures of anomalous volume on the one hand and regulation/compliance measures and real activity measures on the other hand are consistent with the conjecture that anomalous volume is related to exchanges’ incentives to inflate volume.

¹⁷See: <https://www.trustpilot.com/review/coinex.com> for examples of user complaints regarding CoinEx. See complaints about possible manipulation of Bibox and OKX trading volumes at <https://medium.com/@pandahanda44/bibox-artificially-inflated-volume-in-pictures-ebf6cbe1ed2a> and <https://medium.com/@sylvainartplayribes/chasing-fake-volume-a-crypto-plague-ea1a3c1e0b5e>, respectively. See complaints about funds disappearance on Huobi: <https://medium.com/@revblc/hacked-at-huobi-or-an-inside-job-you-decide-7979553cae9f>.

¹⁸The results obtained using number-of-trades-based measures of anomalous trading are qualitatively similar.

Older, more established exchanges tend to exhibit lower levels of anomalous volume measures, consistent with their ability to enjoy network effects and with having a lesser need to resort to information manipulation (e.g., Halaburda and Gandal (2016) and Cong, Miao, Tang, and Xie (2021)). Exchanges with many currencies listed tend to have higher anomalous volume measures. This relation is potentially consistent with product differentiation being negatively related to the importance of reputation (e.g., Kostovetsky and Warner (2016)). In the context of crypto exchanges, an exchange on which more pairs are traded (*ceteris paribus*) has stronger incentives to manipulate trading volume in the short run in a particular currency pair as its future overall profits are less dependent on its reputation among traders in that particular market.

Moving to currency pair characteristics, anomalous volume measures are lower in pairs in which the quote currency is a token. The likely reason is that exchanges have lower incentives to inflate trading volume in pairs involving tokens, since much of the exchanges' compensation in such pairs comes at the listing stage (in the form of listing fee) as opposed to post-listing trading stage. Interestingly, currency pairs listed on an exchange longer tend to exhibit larger anomalous volume measures. A possible reason is the temporal increase in measures of anomalous volume, as evident in Figure 2, coupled with a short sample period, resulting in the oldest exchange-month observations concentrating in the (latest) part of the sample in which aggregate measures of anomalous volume are high.

5.3 Anomalous volume estimates and the Chinese ban

In Table 4, we analyze the effects of a ban on crypto exchanges imposed by regulators in China in 2017 on our measures of anomalous trading. The regulatory action in September 2017 consisted of banning initial coin offerings, restricting any activity that involves crypto tokens, shutdown of exchanges' websites, and canceling exchange business licenses.¹⁹ As a result, several Chinese exchanges, which were responsible at the time for over 90% of Bitcoin trading—Bibox, BigONE, EXX, CoinEx, Gatecoin, Huobi, OkCoin, OKX and ZB—moved headquarters and operations outside of Mainland China,

¹⁹See <https://www.coindesk.com/chinas-ico-ban-a-full-translation-of-regulator-remarks> for details of the regulation.

mainly to Hong Kong and the rest of Pacific Asia.

[Insert Table 4 here]

We use the Chinese ban as an exogenous shock to the ability to inflate trading volume by means of fake trading by (originally) Chinese exchanges. Our conjecture is that moving exchanges overseas presents significant operational challenges that are likely to temporarily reduce exchanges' resources available for volume inflation. If our estimates of anomalous volume are related to fake trading, we expect anomalous volume to decrease for affected exchanges following the ban. To examine the effects of the ban on anomalous volume, we construct a sample in which the treated group includes nine Chinese exchanges that relocated outside of China as a result of the ban. We use two control samples. The first includes all exchanges headquartered outside of China at the time of the ban, according to www.CoinGecko.com and www.Cointelligence.com. The second control sample contains exchanges that were headquartered in Pacific Asia, whose customers are likely to have similar cultural, social, and economic features.²⁰

Our sample period for this analysis includes the first three quarters of 2017 (the “pre-ban period”) and the last quarter of 2017 and first two quarters of 2018 (the “post-ban period”). We estimate regressions of our anomalous volume measures at the exchange-pair-month level on the treated indicator, the post-ban indicator, and the interaction between the two, while controlling for exchange-level and currency-pair-level characteristics, as in Table 3.

The coefficient on Treated is positive and highly economically and statistically significant in five specifications out of six—in these specifications, the level of anomalous volume measures on Chinese exchanges before the the ban was 0.43-0.90 standard deviations higher than on non-Chinese exchanges (in Panel A) and exchanges originally headquartered in Pacific Asia (in Panel B). The coefficients on Post-ban are significant in all specifications, consistent with the general increase over time of anomalous volume measures, as evident in Figure 2. Most importantly, the coefficients on the

²⁰These exchanges are: Binance, Bit-Z, Bitbank, Bitfinex, bitFlyer, BitForex, Bx.in.th, Cobinhood, Ethfinex, HitBTC, KuCoin, LocalBitcoins Quoine, UPbit, and Zaif.

Treated × Post-ban interaction are negative and highly statistically significant in all specifications and are also economically large—Chinese exchanges that moved out of China following the ban reduced the extent of fake trading by 0.19-0.73 standard deviations compared to control exchanges in the three quarters following the ban.

5.4 Anomalous volume estimates and CoinMarketCap index

In Table 5, we examine the evolution of anomalous volume estimates around the introduction by CoinMarketCap—the largest crypto data aggregator—of a new exchange quality index in December 2019.

[Insert Table 5 here]

Prior to that, CoinMarketCap ranked exchanges by their reported volume. Following the index introduction, CoinMarketCap has replaced reported volume on an exchange by a proprietary estimate of exchange quality, which is based in part on estimated volume inflation on the exchange.²¹ If our estimates of anomalous volume are related to fake trading, we expect anomalous trading estimates to decrease more following the liquidity index introduction for exchanges that were most likely to inflate volume prior to the index introduction. This identification strategy relies on differential effects of the index introduction on exchanges with relatively high/low levels of estimated anomalous volume prior to the index introduction. The idea is that the introduction of the index diminishes the importance of trading volume in determination of exchange rankings, lowering exchanges' incentives to manipulate information and inflate volume. Note that this identification strategy is largely independent of whether

²¹CoinMarketCap's Liquidity Score is designed to capture expected slippage (or price impact) of trades: "When executing trades, the most liquid markets have the least slippage," see <https://support.coinmarketcap.com/hc/en-us/articles/360043836931-Liquidity-Score-Market-Pair-Exchange-.18>. CoinMarketCap's methodology for constructing the index thus relies on estimating expected price impact of various trades. CoinMarketCap simulates trades drawn from the range typical of trade sizes by retail traders on centralized exchanges—between \$100 to \$200,000; simulates the price impact of market orders of these trades; and aggregates the results across various order sizes and currency pairs on a particular exchange. The smaller the resulting weighted average price impact the higher the liquidity score.

the index itself is unbiased.²²

We define treated and control exchanges in two ways. First, treated (control) exchanges are those whose rank declines (increases) the most following CoinMarketCap index introduction. The idea is that exchanges that are penalized the most by the index would have reduced incentives to inflate volume.²³ Second, treated exchanges are defined as those with top-five anomalous volume estimates in the three months prior to December 2019 and control exchanges as those with bottom-five anomalous volume estimates during the same period.²⁴ Our sample period is September to November of 2019 (pre-index period) and December, 2019 to February, 2020 (post-index period). The Treated coefficients are significantly positive—treated exchanges have higher anomalous volume. While this result is mechanical in Panel B, where treated exchanges are defined as those with the highest pre-index anomalous volume, the result in Panel A shows that CoinMarketCap methodology is able to identify exchanges with highest anomalous volumes. The Post coefficients are positive as well and are significant in all specifications. This is somewhat surprising, but is consistent with the general trend of temporal increase in anomalous volume, as depicted in Figure 2. Most importantly, the coefficients on the interaction of Treated and Post-index introduction indicators are negative and significant in all specifications. Similar to the case of the Chinese ban, the fact that anomalous volume behaves in a way that is consistent with our conjecture regarding the expected evolution of fake trading provides another piece of evidence consistent with anomalous volume being, at least in part, a result of fake trading.

Overall, the body of evidence presented in the last four subsections suggests that our statistical measures of anomalous volume are likely to be correlated with true, unobservable fake trading, as exchanges that are likely to have lower/reduced incentives/ability to inflate volume exhibit lower/reduced

²²There are reasons to believe that the index may not be unbiased, as the introduction of the index coincided with the acquisition of CoinMarketCap by Binance – the largest crypto exchange.

²³Table B.3 in Appendix B reports exchanges with the largest losses and gains in the ranking following the index introduction.

²⁴Exchanges' estimated anomalous volume during the three months prior to December 2019 are found in Table B.4 in Appendix B.

anomalous volume estimates. Therefore, in what follows, we will explicitly use the terms “volume inflation” and “fake trading” when referring to PCA-based, MAD-based, and EDM-based anomalous volume measures.

6 Competition and fake trading

The previous section showed that our measures of anomalous volume are likely correlated with fake trading, which is a form of information manipulation by crypto exchanges. In this section, to examine the link between information manipulation and competition among exchanges, we study the relation between estimated fake trading (as a type of information manipulation) and measures of competition among exchanges. In particular, we examine the following hypothesis:

Hypothesis 1: An exchange’s incentives to manipulate information through volume inflation are increasing in the extent of competition it is facing.

The intuition is twofold. First, intense competition may increase benefits of manipulating information as a way for less reputable exchanges to gain (legitimate) trading volume. Second, in more competitive markets the likelihood of survival is lower, reducing reputational concerns related to manipulation. We test this hypothesis in two ways—first focusing on the relation between fake trading measures and a static measure of competition among exchanges within a given currency pair, and then examining how fake trading is related to dynamic competition (i.e. to changes in competition).

6.1 Static competition among exchanges and fake trading

In Table 6, we examine associations between anomalous volume estimates and the logarithm of the number of exchanges on which a given currency pair was traded in the month preceding the month of the observation. The larger this number, the stronger the intensity of competition in the market for this particular currency pair. We control for all exchange-level and pair-level characteristics and for

fixed effects, as in Table 3.

[Insert Table 6 here]

Consistent with Hypothesis 1, in Panel A of Table 6 we observe a positive relation between the number of exchanges on which a pair is traded and the average level of fake trading in it. A one-standard-deviation increase in the log of the number of exchanges on which a pair is traded is associated with 0.16-0.34 standard-deviation-increase in volume inflation.

It is possible that the relation between the number of exchanges on which a pair is traded and volume inflation estimates is driven by pairs involving low-cap quote currencies—as such currencies a) dominate trading landscape, b) feature relatively low real trading volume, resulting in exchanges’ low incentives to inflate trading volume in such pairs, and c) are typically traded on few exchanges. Thus, our results may be driven by the spurious relation between volume inflation and the number of exchanges for low-cap pairs, which is not due to competitive effects. To examine this hypothesis, in Panels B and C of Table 6, we focus on two subsets of our data that contain relatively large/important currency pairs.

In Panel B, we examine a subsample of currency pairs in which the quote currency was covered by [www.CoinMarketCap.com](https://www.coinmarketcap.com) in the month preceding the month of the observation. CoinMarketCap tends to cover relatively reputable currencies. In Panel C, we limit our sample to currency pairs for which the market cap of the quote currency exceeded \$U.S. 100 million in the month preceding the observation month. This restriction reduces the sample size by a factor of four. Overall, the results in the last two columns of Table 6 are consistent with the full-sample results—the coefficients on log (number of exchanges) remain positive and significant and have similar magnitudes to the full-sample coefficients. These results support our interpretation of the full-sample results—i.e. that variation in competition among exchanges is at least partially responsible for the variation in fake trading on exchanges.

6.2 Dynamic competition among exchanges and fake trading

The results in the previous subsection suggest that a static measure of competition—the lagged number of exchanges on which a currency pair is traded—is related to volume inflation in that pair. We now examine how dynamic competition among exchanges influences the extent of volume inflation. In Table 7, we examine the effects of entry of rival exchanges in a given currency pair on anomalous volume estimates on exchanges on which this pair has already been traded prior to the entry, while controlling for all exchange-level and pair-level characteristics, as in Table 6, as well as for static competition among exchanges. In particular, we augment the regressions by two variables: competitor entry indicator and competitor entry indicator multiplied by log of the lagged number of exchanges on which the currency pair was traded prior to the entry. Entry of a (competitor) exchange into a given currency pair is a situation in which an exchange that has not previously listed the currency pair reports trading in it in a given month. We define two types of competitors. “General” competitors are a set of all exchanges in our dataset. “Geographical” competitors are a subset of exchanges that are headquartered in the same geographical region as the focal exchange, where regions are defined as in Table B.2 in Appendix B. The geographical dimension of competition is important in the crypto market. Shams (2020) shows that the clientele of investors on crypto exchanges is strongly related to their location.

[Insert Table 7 here]

The main result in Table 7 is that an (additional) exchange listing a currency pair (“competitor entry”) is associated with a 0.17-0.38 standard-deviation increase in volume inflation measures in that pair on exchanges on which the currency pair has already been traded (focal exchanges). The effect tends to be stronger in cases in which entry occurs in an exchange headquartered in the same geographical area as the focal exchange. In addition, the effect of competitor entry tends to be weaker in cases in which the number of exchanges on which the pair has already been traded is high. For example, compared with an entry of an exchange into the market of a pair already traded on 3 exchanges relative to the case of

a pair already traded on 10 exchanges—the effect on volume inflation is 0.06-0.14 standard deviations lower in the latter case. This is consistent with the competitive effect of entry into a new market being stronger in more concentrated markets and being less pronounced in already competitive markets. Overall, the results in Table 7 demonstrate that dynamic evolution of competitive environment leads to changes in anomalous volume estimates in ways that are consistent with competition affecting exchanges’ incentives to manipulate trading volume by means of fake trading.

7 Consequences of fake trading

The results in the previous section suggest that volume inflation may be partially driven by competitive pressure, leading exchanges to inflate trading volume. In this section we examine short-term and long-term effects of fake trading. In particular, we examine the following hypothesis:

Hypothesis 2: Volume inflation by exchanges has positive short-run real effects and negative long-run real effects.

This hypothesis would hold if the traders do not detect volume inflation instantaneously, but do detect it over time. We test this hypothesis by examining the relations between past anomalous volume estimates on one hand and proxies for an exchange’s performance on the other hand. The first proxy for an exchange’s performance is its web popularity, measured by the (log of) the number of visits to the exchange’s website, as reported by www.SimilarWeb.com. The second proxy is the exchange’s social media popularity, measured by the (log of) the number of retweets of tweets initiated by the exchange’s Twitter handle. Web traffic and other measures of user’s activity are often used by as a proxy for the demand by traders.²⁵ While these measures are only indirectly related to an exchange’s performance, they are not controlled by the exchange, and are driven by the interest in the exchange by the public. Thus, these measures are less sensitive to manipulation than trading-volume-based

²⁵See for instance <https://www.theblock.co/linked/107613/web-traffic-crypto-exchanges-may-record-high>.

measures and can serve as unbiased proxies for exchanges' real activity. In contrast, more direct measures, based on trading volumes or estimated revenues may be severely biased due to pervasive volume inflation, which exhibits large cross-sectional and time-series variation.

In Table 8, we examine the effects of past fake trading on an exchange on its web popularity at various horizons.

[Insert Table 8 here]

In particular, we estimate regressions, at the exchange-month level, of an exchange's web popularity on volume inflation measured over periods of three to 24 months prior, while controlling for lagged popularity (as popularity tends to be persistent), exchange's characteristics, currency-pair characteristics, quote currency fixed effects, geographical region fixed effects, and year-month fixed effects.

At the three-month horizon, web popularity increases in measures of anomalous volume—the coefficients on it are positive and significant in all three specifications. Increasing anomalous volume measures by one standard deviation raises the number of visits to the exchange's website by 0.36-0.95 standard deviations. This is consistent with fake trading being effective in the short run. At the six-month horizon, the effect of past volume inflation on exchange's web popularity is somewhat smaller—a one-standard-deviation increase in measures of anomalous volume is associated with 0.26-0.73 standard deviation increase in web popularity. Moving to longer horizons, the relation between past volume inflation and exchange's web popularity turns significantly negative in two specifications out of three at the 12-month horizon. The negative effect is attenuated at the 24-month horizon, although it is still significant in two specifications out of three. This is consistent with reputations effects dissipating with time, especially in as fast-paced environment as the crypto exchange industry.

In Table 9, we perform similar analysis, while using exchange's social media popularity as a proxy for its performance.

[Insert Table 9 here]

At short horizons (three and six months) lagged anomalous volume tends to be positively associated with social media popularity: the coefficients on volume inflation are positive and significant in all specifications. At longer horizons, the relation turns negative: It is significantly negative in two specifications out of three at the 12-month and 24-month horizons. Similar to the case of web popularity, the negative effect of past fake trading on social media popularity fades somewhat in the second year.

Overall, the finding that anomalous volume—which we argue is related to fake trading—has positive short-term effects on the exchange’s popularity at short horizons and negative effects at longer horizons suggests that volume inflation has short-term benefits and long-term costs. This result is consistent with Hypothesis 2, i.e. with exchanges choosing the degree of information manipulation while trading off the near-term increase in operating performance against longer-term adverse reputational effects of information manipulation.

8 Conclusions

We examine the effects of competition among crypto exchanges on their incentives to engage in an opportunistic strategy of inflating (faking) trading volume. The benefit of this strategy is that an exchange with a (seemingly) large trading volume is attractive to potential traders, as larger volume is likely to be associated with lower expected trading costs. The cost of volume inflation is reputational damages, which occur once traders realize that not all reported trading volume is legitimate.

We find a robust relation between competition and the extent of anomalous volume, which we show to be likely related to fake trading. A static measure of competition—the number of exchanges trading in a given currency pair—is positively associated with anomalous volume. Results involving dynamic competition—measured by entry into a given currency pair market by competitor exchanges—show that anomalous volume in a currency pair on incumbent exchanges increases upon competitor exchange entry into that currency pair, suggesting that incentives to manipulate information (fake trades) are increasing in competition intensity.

Our analysis of the effectiveness of fake trading, i.e. its ability to generate short-term rents, demonstrates that exchanges are generally successful in misleading traders in the short run. Fake trading raises exchange’s web popularity and social media popularity—both of which are likely to be correlated with operating and financial performance—over relatively short horizons. However, the strategy of inflating trading volume is costly in the long term. The effects of fake trading on longer-run operating performance tend to be negative: Both web and social media popularity tend to be significantly negatively related to fake trading over longer horizons.

We use the crypto exchange industry as a convenient laboratory for examining the effects of competition on firms’ information manipulation—a research question that extends beyond that particular industry. In addition to the general implications, however, our analysis has particular implications for future regulation of crypto exchanges. Our finding of pervasive volume inflation complements nascent empirical literature on crypto exchanges (e.g., [Amiram et al. \(2022\)](#), [Cong et al. \(2022\)](#), and [Cong et al. \(2023\)](#)) in illustrating the consequences of generally lax regulation and low compliance by exchanges, and in highlighting the importance of regulation of the crypto exchange industry, which is one of the cornerstones of the crypto market.

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Figure 1. Evolution of exchange market concentration. This plot shows the evolution of the Herfindahl index (HHI) across exchanges based on the number of currency pairs. Exchange-pair-based HHI is computed as the sum of the squared number of currency pairs traded on each exchange to the squared sum of all numbers of currency pairs traded on all exchanges. HHI scale is reported on the left-hand side of the plot. The grey bars show the total number of distinct exchanges across the period. The number of exchanges is reported on the right-hand side of the plot. All measures are computed monthly from January 2016 to May 2021.

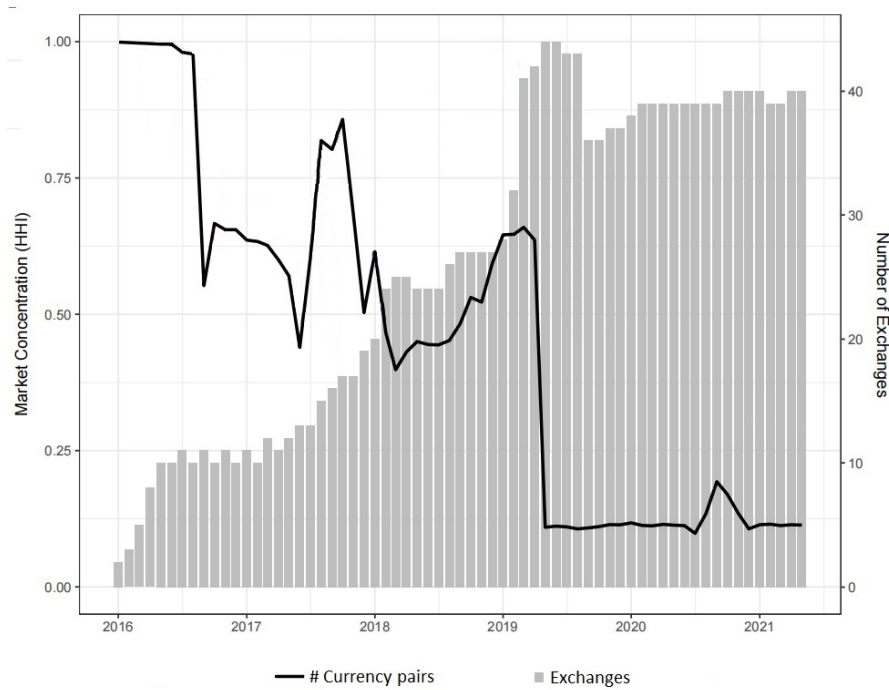
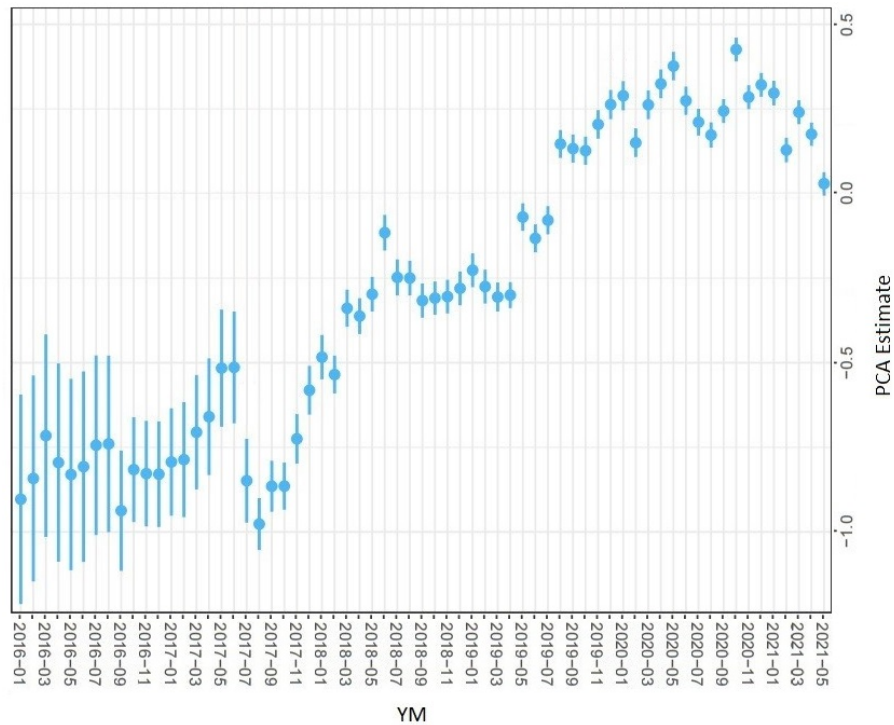


Figure 2. Mean PCA-based measure of anomalous trading over time. This plot shows equally-weighted (across currency pairs and exchanges) means for each month of volume-based PCA-based measure of anomalous trading for the period between January 2016 to May 2021. The lines around point estimates indicate confidence intervals.



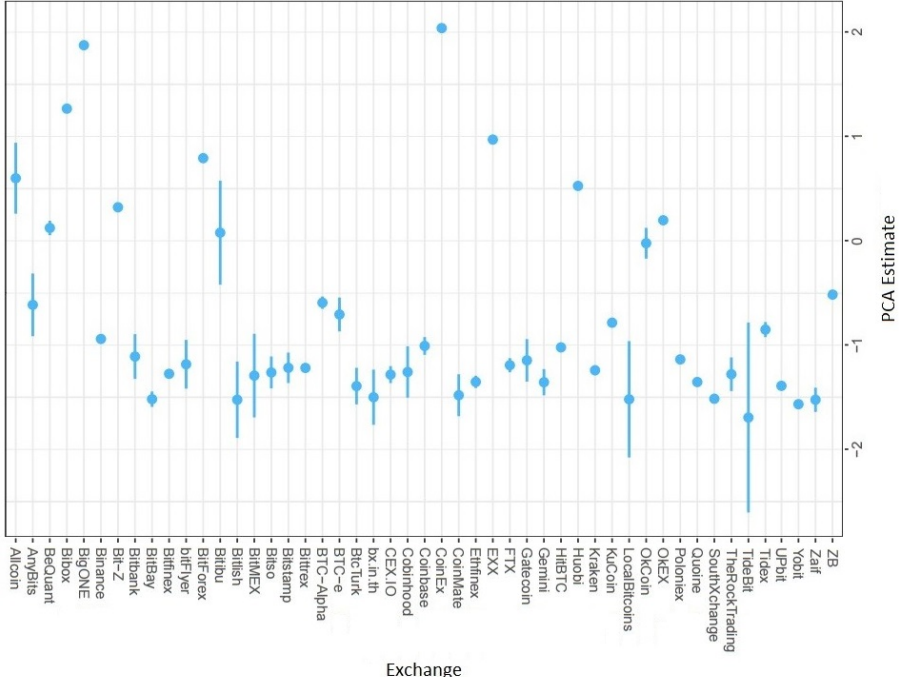


Table 1. Summary statistics. This table reports summary statistics of the market-level (in Panel A), exchange-level (in Panel B), and currency-pair level (in Panel C) variables used in the empirical analysis. The sample period is May 2013 – May 2021. All variables are defined in Table B.1 in the Appendix.

Panel A: Market						
	Min	Max	Median	Mean	St. Dev.	Obs.
Currencies	1	3,112	236	944.04	967.13	97
Currencies: Entry	0	903	32	84.70	159.05	97
Currencies: Exit	0	941	2	48.35	147.12	97
Currency pairs	1	4,389	255	1,390.34	1,450.60	97
Currency pairs: Entry	0	1,224	62	146.28	224.13	97
Currency pairs: Exit	0	1,957	6	64.59	204.26	97
Volume: USD (MM)	0.00	9,250.75	1.71	340.27	1,021.67	97
# Trades (M)	0.00	3,754.56	24.08	404.11	690.27	97
Exchanges	1	44	12	18.22	15.57	97
Exchanges: Entry	0	1	0	0.47	0.32	97
Exchanges: Exit	0	1	0	0.04	0.20	97
HHI Exchanges: # Currencies	0.10	1.00	0.65	0.63	0.36	97

Panel B: Exchanges						
	Min	Max	Median	Mean	St. Dev.	Obs.
Age	1	97	21	24.34	18.52	1,767
Currency pairs	1	2,433	33	137.39	231.10	1,767
Currency pairs: Entry	0	655	1	9.44	39.44	1,767
Currency pairs: Exit	0	1,873	0	4.96	52.52	1,767
AML	0	1	1	0.51	0.50	1,767
KYC	0	1	1	0.65	0.48	1,767
Traffic (MM)	0.00	6.99	0.04	0.29	0.80	1,697
Twitter retweets	1.00	688.00	26.00	53.41	73.86	782
Github commits	1.00	302.00	7.00	15.00	27.51	412

Panel C: Currency Pairs						
	Min	Max	Median	Mean	St. Dev.	Obs.
Listed on # exchanges	1.00	37.00	2.00	4.18	5.20	237,205
Log (listed on # exchanges)	0.00	3.61	0.89	1.61	0.95	237,205
Age of listing on any exchange	0.00	96.00	20.00	23.87	18.85	237,205
Age of listing on a given exchange	0.00	96.00	12.00	14.51	12.16	237,205
Token	0.00	1.00	1.00	0.53	0.50	237,205

Table 2. Anomalous volume measures and principal components. Panel A reports summary statistics of volume-based and ten-minute-aggregated number-of-trades-based measures of fake trading (MAD, and EDM). Panel B reports these measures normalized to have zero mean and unit standard deviation. Panel C reports summary statistics of normalized principal components based on volume-based measures of anomalous trading, number-of-trades-based measures of anomalous trading, and combined volume-based and number-of-trades-based measures of anomalous trading. Anomalous trading measures and principal components are at the exchange-pair-month level. All measures are described briefly in Sections 4.1, 4.2, and 4.3, described in greater detail in Appendix A, and defined formally in Table B.1 in Appendix B.

Panel A: Original measures						
	Min	Max	Median	Mean	St. Dev.	Obs.
MAD: Volume	0.21	246.00	7.98	21.57	32.56	197,473
MAD: # Trades	0.27	260.00	13.50	36.38	47.10	197,473
EDM: Volume	0.00	594.00	5.00	23.97	56.30	236,324
EDM: # Trades	0.00	585.00	12.00	37.06	60.71	236,324
Panel B: Normalized measures						
	Min	Max	Median	Mean	St. Dev.	Obs.
MAD: Volume (Norm.)	-0.66	6.89	-0.42	0.00	1.00	197,473
MAD: # Trades (Norm.)	-0.77	4.75	-0.49	0.00	1.00	197,473
EDM: Volume (Norm.)	-0.43	10.13	-0.34	0.00	1.00	236,324
EDM: # Trades (Norm.)	-0.53	7.86	-0.36	0.00	1.00	236,324
Panel C: Normalized principal components						
	Min	Max	Median	Mean	St. Dev.	Obs.
PCA: Volume	-2.88	8.79	-0.44	0.00	1.00	197,470
PCA: # Trades	-5.71	7.68	-0.44	0.00	1.00	197,470
PCA: Both	-5.27	11.5	-0.62	0.00	1.00	197,470

Table 3. Anomalous volume measures and exchange and currency-pair characteristics. This table reports multi-variate regressions of anomalous volume on exchange characteristics. In the first (second, third) column, the dependent variable is PCA (MAD, EDM) normalized measure of anomalous volume. See Table B.1 in Appendix B for variable definitions. The sample contains all exchange-currency pair-month observations with non-missing values of dependent and independent variables. The regressions are estimated at the exchange-pair-month level. All specifications include quote currency, geographical region, and year-month fixed effects. Standard errors (reported in parentheses) are heteroskedasticity consistent and clustered at the exchange $\times$ currency pair level. * Significant at 10 percent; ** Significant at 5 percent; *** Significant at 1 percent.

	PCA	MAD	EDM
Compliance:			
AML	-0.65 *** (0.04)	-0.25 *** (0.02)	-0.83 *** (0.03)
KYC	-0.41 *** (0.03)	-0.10 *** (0.02)	-0.12 *** (0.03)
Real activity:			
log (Web Traffic)	-0.10 *** (0.01)	-0.04 *** (0.00)	-0.01 ** (0.00)
log (Social Media: Tweets)	-0.05 *** (0.00)	-0.06 *** (0.00)	-0.06 *** (0.00)
log (Code Activity: Commits)	-0.09 *** (0.01)	-0.07 *** (0.00)	-0.05 *** (0.01)
Additional exchange characteristics:			
log (Age)	0.01 (0.02)	-0.11 *** (0.01)	-0.31 *** (0.02)
log (# Currency pairs)	0.11 *** (0.01)	0.13 *** (0.01)	0.02 * (0.01)
Currency pair characteristics:			
Token	-0.04 (0.02)	-0.05 * (0.02)	0.04 (0.03)
log (Age of listing on exchange)	0.09 *** (0.01)	0.07 *** (0.01)	0.09 *** (0.02)
Quote FE	yes	yes	yes
Geo Region FE	yes	yes	yes
Year-month FE	yes	yes	yes
Obs.	196,892	196,895	235,559
Adj. r^2	0.33	0.37	0.28

Table 4. Exogenous shock: the Chinese ban. This table reports regression estimates of anomalous volume around the Chinese ban on crypto exchanges in September 2017. The sample period includes the first three quarters of 2017 (the pre-ban period) and the last quarter of 2017 and the first two quarters of 2018 (the post-ban period). In Panel A, Treated equals one (zero) for exchanges headquartered in China (outside of China). In Panel B, Treated equals one (zero) for exchanges headquartered in China (Pacific Asia). Post equals one (zero) in the post-ban (pre-ban) period. Treated $\times$ Post is the product of the two indicators. In the first (second, third) column of each of the panels, the dependent variable is PCA (MAD, EDM) normalized measure of anomalous volume. See Table B.1 in Appendix B for variable definitions. The regressions are estimated at the exchange-pair-month level. Standard errors (reported in parentheses) are heteroskedasticity consistent and clustered at the currency pair level. * Significant at 10 percent; ** Significant at 5 percent; *** Significant at 1 percent.

	Panel A: Outside China			Panel B: Pacific Asia		
	PCA	MAD	EDM	PCA	MAD	EDM
Indicators of Interest:						
Treated	0.90 *** (0.05)	0.86 *** (1.13)	0.43 *** (1.80)	0.57 *** (0.08)	0.73 *** (1.69)	-0.01 (2.56)
Post	0.19 *** (0.01)	0.24 *** (0.01)	0.48 *** (0.02)	0.30 *** (0.03)	0.31 *** (0.03)	0.56 *** (0.04)
Treated $\times$ Post	-0.33 *** (0.03)	-0.19 *** (0.03)	-0.29 *** (0.04)	-0.73 *** (0.07)	-0.40 *** (0.06)	-0.31 *** (0.08)
Exchange Controls	yes	yes	yes	yes	yes	yes
Currency Pair Controls	yes	yes	yes	yes	yes	yes
Quote FE	yes	yes	yes	yes	yes	yes
Obs.	26,177	26,178	33,296	11,697	11,698	12,139
Adj. r^2	0.33	0.41	0.36	0.14	0.14	0.10

Table 5. Exogenous shock: CoinMarketCap index introduction. This table reports regression estimates of anomalous volume around the CoinMarketCap index introduction in December 2019. The sample period includes September to November 2019 (the pre-introduction period) and December 2019 to February 2020 (the post-introduction period). In Panel A, Treated equals one (zero) for exchange with top-five (bottom-five) reduction in CoinMarketCap ranking in the three months after the index introduction relative to three months prior to it (see Table B.3 in Appendix B). In Panel B, Treated equals one (zero) for exchange with top-five (bottom-five) anomalous volume estimates during the three months before the index introduction (see Table B.4 in Appendix B). Post equals one (zero) in the post-introduction (pre-introduction) period. Treated $\times$ Post is the product of the two indicators. In the first (second, third) column of each of the panels, the dependent variable is PCA (MAD, EDM) normalized measure of anomalous volume. See Table B.1 in Appendix B for variable definitions. The regressions are estimated at the exchange-pair-month level. All specifications include quote currency, geographical region, and year-month fixed effects. Standard errors (reported in parentheses) are heteroskedasticity consistent and clustered at the currency pair level. * Significant at 10 percent; ** Significant at 5 percent; *** Significant at 1 percent.

	Panel A: Change in CMC ranking			Panel B: Pre-index volume inflation		
	PCA	MAD	EDM	PCA	MAD	EDM
Indicators of Interest:						
Treated	0.55 *** (0.08)	0.10 (0.08)	0.15 *** (0.04)	2.51 *** (0.14)	2.01 *** (0.10)	0.86 *** (0.15)
Post	0.48 *** (0.04)	0.64 *** (0.05)	0.19 *** (0.01)	0.29 *** (0.06)	0.13 *** (0.05)	0.23 *** (0.06)
Treated $\times$ Post	-0.50 *** (0.05)	-0.69 *** (0.06)	-0.14 *** (0.02)	-0.15 ** (0.06)	-0.10 *** (0.04)	-0.28 *** (0.06)
Exchange Controls	yes	yes	yes	yes	yes	yes
Currency Pair Controls	yes	yes	yes	yes	yes	yes
Quote FE	yes	yes	yes	yes	yes	yes
Obs.	10,694	11,265	10,694	11,697	11,698	12,139
Adj. r^2	0.44	0.22	0.33	0.14	0.15	0.11

Table 6. Anomalous volume and static competition. This table reports estimates of regressions of anomalous volume measures on static competition at the exchange-pair-month level, measured by the the number of exchanges on which a currency pair is listed in a given month. In columns 1, 4, and 7 the dependent variable is PCA-based measure. In columns 2, 5, and 8 the dependent variable is MAD-based measure. In columns 3, 6, and 9 the dependent variable is EDM-based measure. Panel A reports the regression results on the entire sample. In Panel B, the sample is limited to pairs covered by www.CoinMarketCap.com. In Panel C, the sample is limited to pairs involving quote assets with market capitalization exceeding \$US 100M in the preceding month. See Table B.1 in Appendix B for variable definitions. The regressions are estimated at the exchange-pair-month level. Standard errors (reported in parentheses) are heteroskedasticity consistent and clustered at the currency pair level. * Significant at 10 percent; ** Significant at 5 percent; *** Significant at 1 percent.

	PCA	MAD	EDM	PCA	MAD	EDM	PCA	MAD	EDM
	Panel A: Main Specification			Panel B: CMC Covered			Panel C: Large Caps		
Static competition:									
log (Lagged listed on # Exchanges)	0.18 *** (0.01)	0.16 *** (0.01)	0.34 *** (0.02)	0.15 *** (0.02)	0.12 *** (0.01)	0.30 *** (0.02)	0.11 *** (0.03)	0.11 *** (0.02)	0.24 *** (0.04)
Real activity:									
log (Web Traffic)	-0.11 *** (0.01)	-0.05 *** (0.00)	-0.03 *** (0.00)	-0.13 *** (0.01)	-0.05 *** (0.01)	-0.05 *** (0.01)	-0.17 *** (0.01)	-0.07 *** (0.01)	-0.10 *** (0.01)
log (Social Media: Tweets)	-0.05 *** (0.01)	-0.06 *** (0.00)	-0.06 *** (0.00)	-0.04 *** (0.01)	-0.04 *** (0.01)	-0.06 *** (0.01)	-0.03 *** (0.01)	-0.01 * (0.01)	-0.07 *** (0.01)
log (Code Activity: Commits)	-0.09 *** (0.01)	-0.07 *** (0.00)	-0.05 *** (0.01)	-0.10 *** (0.01)	-0.08 *** (0.01)	-0.06 *** (0.01)	-0.15 *** (0.01)	-0.09 *** (0.01)	-0.07 *** (0.01)
Exchange characteristics	yes	yes	yes	yes	yes	yes	yes	yes	yes
Currency-pair characteristics	yes	yes	yes	yes	yes	yes	yes	yes	yes
Quote FE	yes	yes	yes	yes	yes	yes	yes	yes	yes
Geo Region FE	yes	yes	yes	yes	yes	yes	yes	yes	yes
Year-month FE	yes	yes	yes	yes	yes	yes	yes	yes	yes
Obs.	186,116	186,118	221,585	134,057	134,059	144,539	50,528	50,528	51,583
Adj. r ²	0.35	0.38	0.30	0.35	0.37	0.20	0.33	0.30	0.16

Table 7. Anomalous volume and dynamic competition. This table reports estimates of regressions of anomalous volume measures on dynamic competition at the exchange-pair-month level, measured by competitor entry, alone and interacted with the logarithm of the lagged number of exchanges on which a currency pair was traded in previous month. Competitor entry is an indicator equaling one if there exists at least one exchange that has not previously listed a currency pair starts listing it in a given month. In columns 1, and 4, the dependent variable is PCA-based measure of anomalous trading. In columns 2 and 5, the dependent variable is MAD-based measure. In columns 3 and 6, the dependent variable is EDM-based measure. In Panel A, “general” competitors are a set of all exchanges in our dataset. In columns 4-6, “geographical” competitors are a subset of exchanges that are headquartered in the same geographical region as the focal exchange. See Table B.1 in Appendix B for variable definitions. The regressions are estimated at the exchange-pair-month level. Standard errors (reported in parentheses) are heteroskedasticity consistent and clustered at the currency pair level. * Significant at 10 percent; ** Significant at 5 percent; *** Significant at 1 percent.

	PCA	MAD	EDM	PCA	MAD	EDM
	Panel A: General			Panel B: Geographical		
Dynamic competition:						
Competitor entry × log (Lagged listed on # exchanges)	-0.06 *** (0.02)	-0.06 *** (0.02)	-0.07 *** (0.02)	-0.10 *** (0.03)	-0.14 *** (0.02)	-0.12 *** (0.02)
Competitor entry	0.20 *** (0.04)	0.17 *** (0.03)	0.26 *** (0.04)	0.29 *** (0.04)	0.38 *** (0.04)	0.37 *** (0.04)
Static competition:						
log (Lagged listed on # exchanges)	0.18 *** (0.01)	0.15 *** (0.01)	0.33 *** (0.02)	0.22 *** (0.01)	0.20 *** (0.01)	0.37 *** (0.02)
Real activity:						
log (Web Traffic)	-0.09 *** (0.01)	-0.01 (0.00)	-0.01 * (0.01)	-0.11 *** (0.01)	-0.05 *** (0.00)	-0.03 *** (0.00)
log (Social Media: Tweets)	-0.08 *** (0.01)	-0.05 *** (0.00)	0.00 (0.00)	-0.05 *** (0.01)	-0.06 *** (0.00)	-0.06 *** (0.00)
log (Code Activity: Commits)	-0.12 *** (0.01)	-0.04 *** (0.01)	0.01 (0.01)	-0.09 *** (0.01)	-0.07 *** (0.00)	-0.05 *** (0.01)
Exchange characteristics	yes	yes	yes	yes	yes	yes
Currency-pair characteristics	yes	yes	yes	yes	yes	yes
Quote FE	yes	yes	yes	yes	yes	yes
Geo Region FE	yes	yes	yes	yes	yes	yes
Year-month FE	yes	yes	yes	yes	yes	yes
Obs.	186,116	186,118	221,585	186,116	186,118	221,585
Adj. r ²	0.35	0.38	0.30	0.23	0.23	0.26

Table 8. Real effects of fake trading on web popularity. This table reports estimates of regressions of a measure of web popularity on anomalous volume estimates over the past 3 (6, 12, 24) months, in Panel A (B, C, D). Web popularity is measured as the log of monthly web traffic to the exchange. In columns 1 and 4, the anomalous volume measure is PCA. In columns 2 and 5, the anomalous volume measure is MAD. In columns 3 and 6, the anomalous volume measure is EDM. We control for lagged web popularity, all exchange and currency-pair characteristics, described in detail in Table 6 and defined in Table B.1 in Appendix B. All specifications include quote currency, geographical region, and year-month fixed effects. The regressions are estimated at the exchange-month level. Standard errors (reported in parentheses) are heteroskedasticity consistent. * Significant at 10 percent; ** Significant at 5 percent; *** Significant at 1 percent.

	PCA	MAD	EDM	PCA	MAD	EDM
	Panel A - 3 Months			Panel C - 12 Months		
log (Lagged anomalous volume)	0.95 *** (0.14)	0.36 *** (0.08)	0.43 *** (0.07)	-0.36 ** (0.15)	-0.09 (0.09)	-0.19 ** (0.08)
Lagged web popularity	yes	yes	yes	yes	yes	yes
Exchange characteristics	yes	yes	yes	yes	yes	yes
Currency-pair characteristics	yes	yes	yes	yes	yes	yes
Quote FE	yes	yes	yes	yes	yes	yes
Geo Region FE	yes	yes	yes	yes	yes	yes
Year-month FE	yes	yes	yes	yes	yes	yes
Obs.	1,635	1,635	1,635	1,221	1,221	1,221
Adj. r ²	0.42	0.41	0.42	0.37	0.37	0.37
	Panel B - 6 Months			Panel D - 24 Months		
log (Lagged anomalous volume)	0.79 *** (0.14)	0.26 *** (0.08)	0.36 *** (0.07)	-0.07 *** (0.03)	-0.03 (0.08)	-0.14 ** (0.07)
Lagged web popularity	yes	yes	yes	yes	yes	yes
Exchange characteristics	yes	yes	yes	yes	yes	yes
Currency-pair characteristics	yes	yes	yes	yes	yes	yes
Quote FE	yes	yes	yes	yes	yes	yes
Geo Region FE	yes	yes	yes	yes	yes	yes
Year-month FE	yes	yes	yes	yes	yes	yes
Obs.	1,489	1,489	1,489	1,113	1,113	1,113
Adj. r ²	0.41	0.40	0.40	0.36	0.35	0.36

Table 9. Real effects of fake trading on social media popularity. This table reports estimates of regressions of a measure of social media popularity on anomalous volume estimates over the past 3 (6, 12, 24) months, in Panel A (B, C, D). Social media popularity is measured as log of monthly retweets of tweets from the exchange's handle. In columns 1 and 4, the anomalous volume measure is PCA. In columns 2 and 5, the anomalous volume measure is MAD. In columns 3 and 6, the anomalous volume measure is EDM. We control for lagged social media popularity, all exchange and currency-pair characteristics, described in detail in Table 6 and defined in Table B.1 in Appendix B. All specifications include quote currency, geographical region, and year-month fixed effects. The regressions are estimated at the exchange-month level. Standard errors (reported in parentheses) are heteroskedasticity consistent. * Significant at 10 percent; ** Significant at 5 percent; *** Significant at 1 percent.

	PCA	MAD	EDM	PCA	MAD	EDM
	Panel A - 3 Months			Panel C - 12 Months		
log (Lagged anomalous volume)	0.13 ** (0.06)	0.12 *** (0.03)	0.16 *** (0.03)	-0.20 ** (0.08)	-0.03 (0.05)	-0.12 *** (0.04)
Lagged social media popularity	yes	yes	yes	yes	yes	yes
Exchange characteristics	yes	yes	yes	yes	yes	yes
Currency-pair characteristics	yes	yes	yes	yes	yes	yes
Quote FE	yes	yes	yes	yes	yes	yes
Geo Region FE	yes	yes	yes	yes	yes	yes
Year-month FE	yes	yes	yes	yes	yes	yes
Obs.	1,699	1,699	1,699	1,251	1,251	1,251
Adj. r ²	0.61	0.62	0.62	0.36	0.36	0.36
	Panel B - 6 Months			Panel D - 24 Months		
log (Lagged anomalous volume)	0.21 *** (0.06)	0.12 *** (0.03)	0.18 *** (0.03)	-0.06 *** (0.01)	-0.04 (0.04)	-0.09 ** (0.04)
Lagged social media popularity	yes	yes	yes	yes	yes	yes
Exchange characteristics	yes	yes	yes	yes	yes	yes
Currency-pair characteristics	yes	yes	yes	yes	yes	yes
Quote FE	yes	yes	yes	yes	yes	yes
Geo Region FE	yes	yes	yes	yes	yes	yes
Year-month FE	yes	yes	yes	yes	yes	yes
Obs.	1,534	1,534	1,534	1,146	1,146	1,146
Adj. r ²	0.53	0.53	0.53	0.30	0.30	0.30

Appendix A : Methods for detecting anomalous trading: Detailed descriptions

Examples of anomalous trading

Figure A.1 presents examples of trading activity that illustrate various patterns of anomalous trading.

[Insert Figure A.1 here]

Panel A depicts the evolution of trading volume (in the upper two figures) and ten-minute-aggregated number of trades (in the lower two figures) in the OmiseGo-Bitcoin pair (OMG-BTC) during 4,464 ten-minute intervals in January 2019 on two exchanges—OKX and Binance. The upper two figures suggest that while trading volume on Binance follows an arguably random pattern, there seem to be several structural breaks in trading volume on OKX. Similar structural breaks are evident in the number of trades on OKX, whereas they do not seem to be present on Binance.

Panel B presents trading volume and number of trades in one of the most important cryptocurrency pairs—ETH-BTC—on two exchanges: ZB (on the left) and Binance (on the right) in March 2019. While the plots of both trading volume and number of trades on Binance do not reveal unusual patterns, there are numerous spikes of trading activity on ZB, which are often two orders of magnitude larger than the typical trading volume and occur with a roughly constant frequency.

Benford-Law-based anomalous trading measure

Construction of Benford-Law-based MAD

Benford-Law-Based MAD (mean absolute deviation) statistic is calculated as the sum of the absolute difference between the empirical frequency of each digit, from 1 to 9, and the theoretical frequency found in Benford's Law, divided by the number of leading digits used (9). The construction of our Benfords' distance measure (Benford (1938)) follows Cong et al. (2022). However, we use a different conformity statistic. While Cong et al. (2022) compare the differences in first-significant-digit distribution and Benford's Law at a particular moment in time using Chi-squared statistic, we use the mean-absolute-distance (MAD) of the differences in first-significant-digit distribution and Benford's Law across time. The reason for our choice is that ours is a panel data; and MAD has been used in the context of Benford-Law-based measure within panel data (e.g., Amiram et al. (2015)).

Illustrations of Benford-Law-based measures

Panel A of Figure A.2 shows examples of observed series of first digits of trading volume over the course of one month (March 2019) in the ETH-BTC pair on four crypto exchanges—Binance, ZB, OKX, and Bibox.

[Insert Figure A.2 here]

In each of the four figures, the solid line depicts the actual distribution of leading digits of trading volume, whereas the dashed line depicts theoretical, Benford-Law-based distribution. The observed series for Binance seems to roughly approximate the theoretical Benford’s Law. However, the other plots indicate frequencies of leading digits that deviate substantially from Benford’s Law. For instance, the plot for ZB shows that the frequencies of digits 7-9 are over twice as large as those predicted by Benford’s Law; the same is true for digit 5 on OKX and digits 2 and 3 on Bibox. The deviations from Benford’s Law are even more striking in Panel B of Figure A.2, which presents examples of frequencies of first digits of ten-minute-aggregated number of trades series.

E-divisive with medians (EDM)-based anomalous trading measure

Construction of EDM measure

EDM recursively partitions a time-series and uses a permutation test to determine change points. This non-parametric method has at least three advantages over parametric ones. First, EDM can detect both “mean shift” (sudden change) and “ramping” (gradual change) in data series. This seems especially relevant in our case, as illustrated in Figure A.1. Second, EDM uses moving median as opposed to mean, and, as a result, is robust to the presence of anomalies, i.e. irregular large trades. Finally, the non-parametric nature of EDM implies that the model adapts to the underlying distribution in the data.

We use the Twitter BreakoutDetection package for R as an implementation of the EDM algorithm.^{26,27} To distinguish a short-term tick up or down from a change in the underlying signal, this package decomposes a time-series into a series of segments of one of three types: (a) steady state: the-time

²⁶See <https://github.com/twitter/BreakoutDetection>.

²⁷The model is set at its standard form with the following parameters: method=‘multi’, min.size = 6, beta=0.002, and degree=1. Adjusting beta down (up) makes the EDM estimate more (less) sensitive to breakouts in the distribution. For an example, see https://blog.twitter.com/engineering/en_us/a/2014/breakout-detection-in-the-wild. We set beta at .002 following standard practice in the computer science literature (e.g., James and Matteson (2015), and James et al. (2016)).

series follows a fixed mean (with random noise around the mean); (b) mean shift: the time-series jumps directly from one steady state to another; and (c) ramp up/down: the time-series transitions linearly from one steady state to another, over a fixed period. The breakout algorithm then returns a list of breakout points, or in other words, the number of times these state transitions are detected.

Measurement of EDM-based anomalous trading

Figure A.3 presents two examples of the performance of EDM algorithm in detecting structural breaks in our data.

[Insert Figure A.3 here]

The first example, presented in Panel A, is number of trades in the OMG-BTC pair on the OKX exchange in January 2019. The series is characterised by a step-like shape. The EDM algorithm is able to correctly detect four structural breaks, highlighted by dashed vertical lines (where the timing of the breaks is identified by the algorithm). Importantly, due to its reliance on comparisons of medians (and not means), the algorithm also correctly ignores a temporary spike in trading around bin 3,800. The second example, depicted in Panel B, is trading volume in the ETH-BTC pair on Binance in May 2019. The EDM algorithm does not identify any breakouts in this case, consistent with no visible structural breaks in the trading volume plot.

Principal-components-based measure of anomalous trading

In constructing our principal components measures of anomalous trading, we are guided by [Kaiser \(1961\)](#), [Kaiser \(1970\)](#), [Jolliffe \(2002\)](#), [Peres-Neto and Somers \(2005\)](#), and [Abdi and Williams \(2010\)](#), who propose tests of the appropriateness of factor analysis based on the variance of the first principal components, correlations with underlying measures, and vector direction. All original measures of fake trade pass Bartlett's test of sphericity²⁸ and Kaiser-Meyer-Olkin test.²⁹

²⁸Bartlett's sphericity test provides information about whether the correlations in the data are strong enough to use a dimension-reduction technique such as principal components. The test asks whether a correlation matrix is an identity matrix. If so, the variables are uncorrelated, and one cannot perform a principal components analysis to reduce the dimensionality of the data. More formally, Bartlett's sphericity test is a test of whether the data are a random sample from a multivariate normal distribution, the covariance matrix of which is a diagonal matrix. In our case, the test statistic is very high (38.16), leading to a rejection of the null hypothesis and supporting the use of a dimension-reduction technique (PCA).

²⁹Kaiser-Meyer-Olkin (KMO) test is a measure of how suited the data are for factor analysis. The test measures sampling adequacy for each variable in the model and for the model as a whole. The statistic (ranging between zero

Panel A of Figure A.4, which presents a biplot of the orthogonalization of MAD and EDM measures and their relation with the first two principal components for both volume-based and number-of-trades-based measures. The length of each vector (squared cosine) represents the respective variable in the first two principal components. The horizontal and vertical projections of a vector represent the variable in the first and second principal components, respectively. Panel B of the same figure, which presents correlations between the first and second principal components of MAD and EDM. Both Panel A and Panel B show that these correlations are quite high but far from perfect—ranging between 0.6 and 0.777, suggesting that there is additional information in principal-component-based measures.

[Insert Figure A.4 here]

Panel C of Figure A.4 presents fractions of variance explained by the first four principal components of the individual anomalous trading measures. Since the first principal component explains a significant fraction of variation in anomalous trading—56%, we concentrate on the first principal component as an aggregated measure of anomalous trading in the empirical analysis, in addition to MAD and EDM measures.

and one) is a measure of the fraction of variance among variables that is common. The higher the proportion, the more suited the data are to factor analysis. As rule of thumb, factor analysis is appropriate for values between 0.8 and 1. In our sample, the statistic equals 0.871. This indicates significant overlap of information among our measures of anomalous volume supports the use of factor analysis.

Figure A.1. Examples of trading volume and number of trades series. The plot shows trading volume (in upper figures of Panels A and B) and number of trades (in the lower figures of Panels A and B), aggregated into 4,464 ten-minute bins throughout a one-month period. Panel A displays data for OMG-BTC pair on OKX (left panels) and Binance (right panels) in January 2019. Panel B displays data for ETH-BTC pair on ZB (left panels) and Binance (right panels) in March 2019.

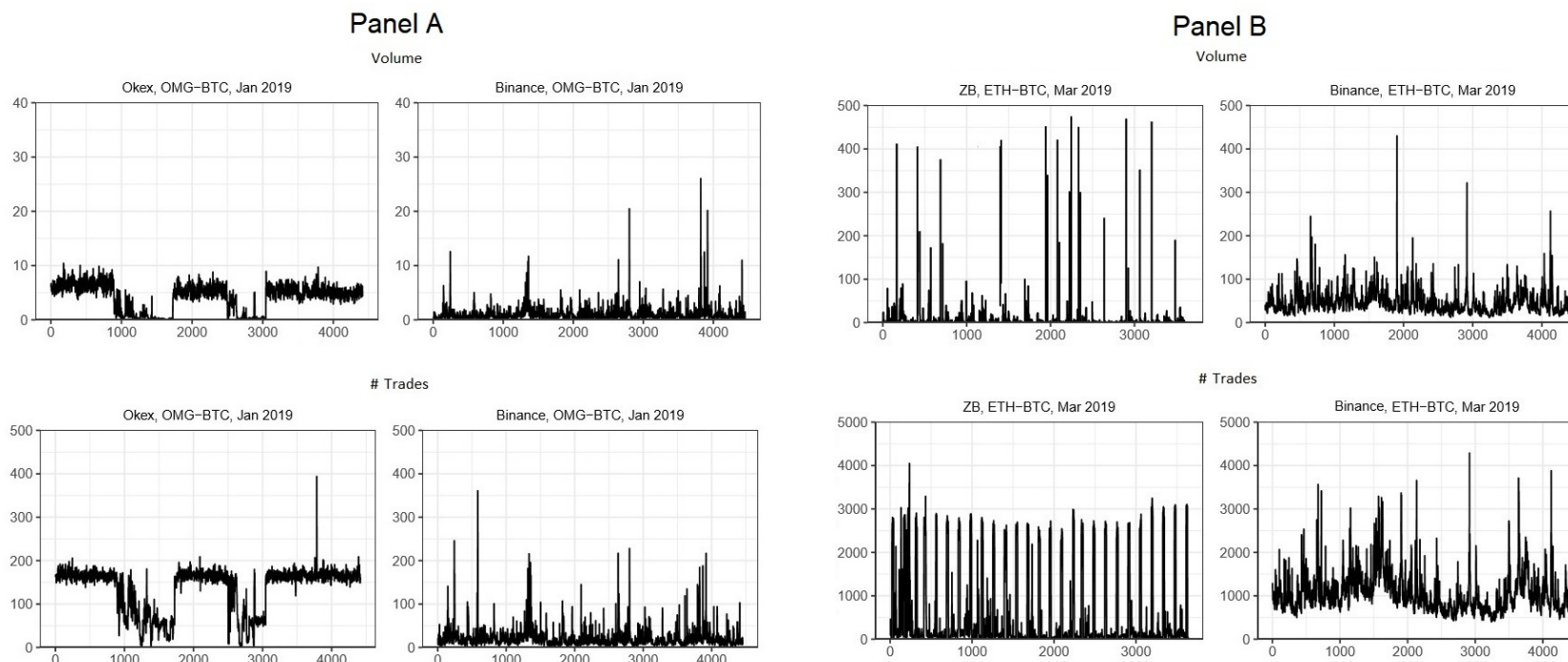


Figure A.2. Examples of deviations from Benford's Law. Blue solid curves represent empirical frequencies of leading digits of trading volume (in Panel A) and ten-minute aggregates of number of trades (in Panel B). Red dotted curves represent frequencies of leading digits under Benford's Law. The frequencies are computed for ETH-BTC pair in March 2019 on four exchanges: Binance, ZB, OKX, and Bibox.

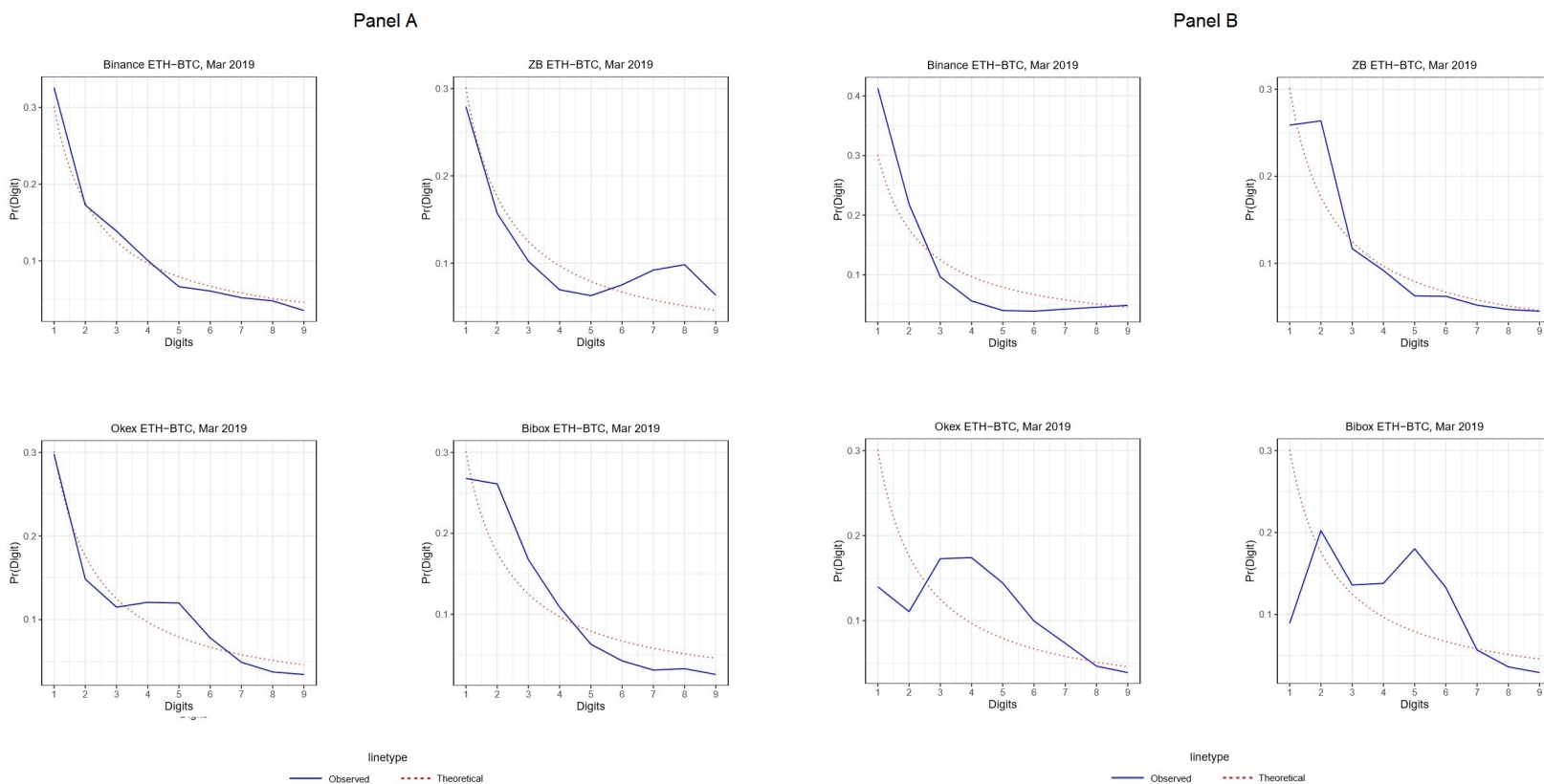


Figure A.3. Examples of application of EDM measure. The plots in both panels represent the number of trades series, aggregated into 4,464 ten-minutes bins. Panel A presents trading in OMG-BTC pair in January 2019 on OKX. Panel B presents trading in ETH-BTC pair on Binance in May 2019. Dashed red lines depicts structural breaks as detected by the EDM algorithm.

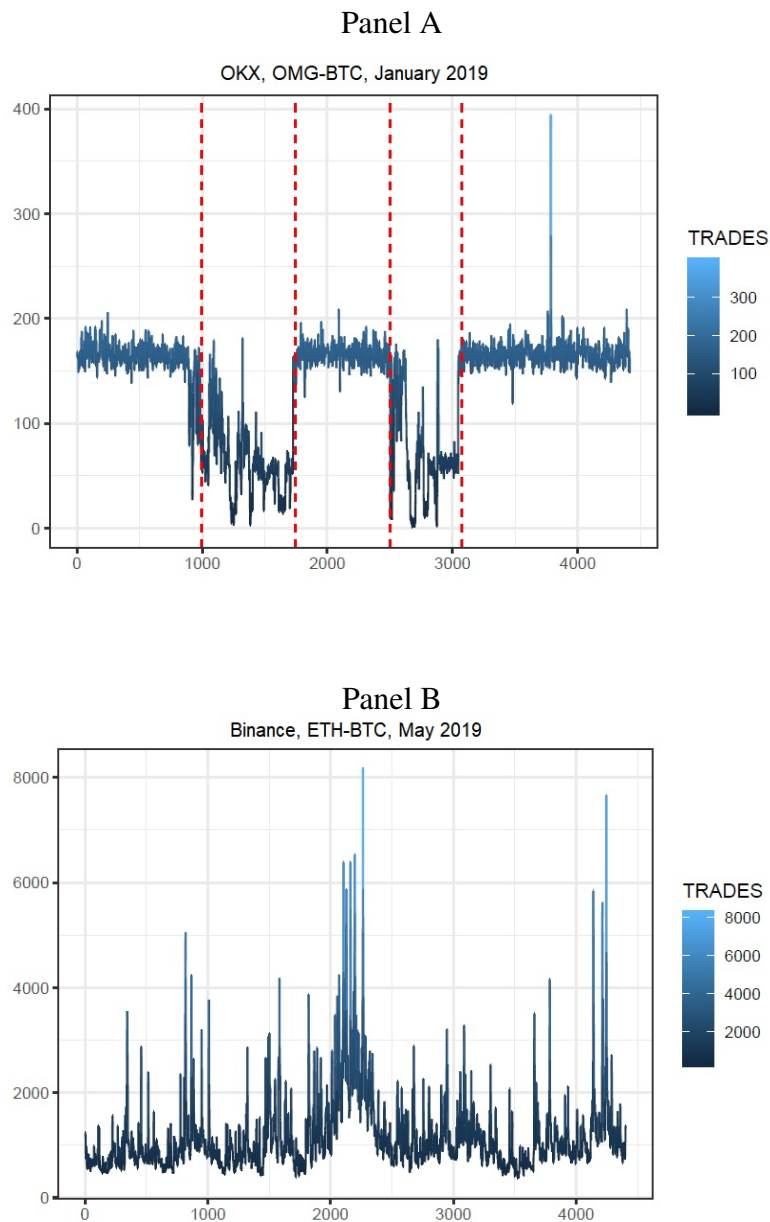
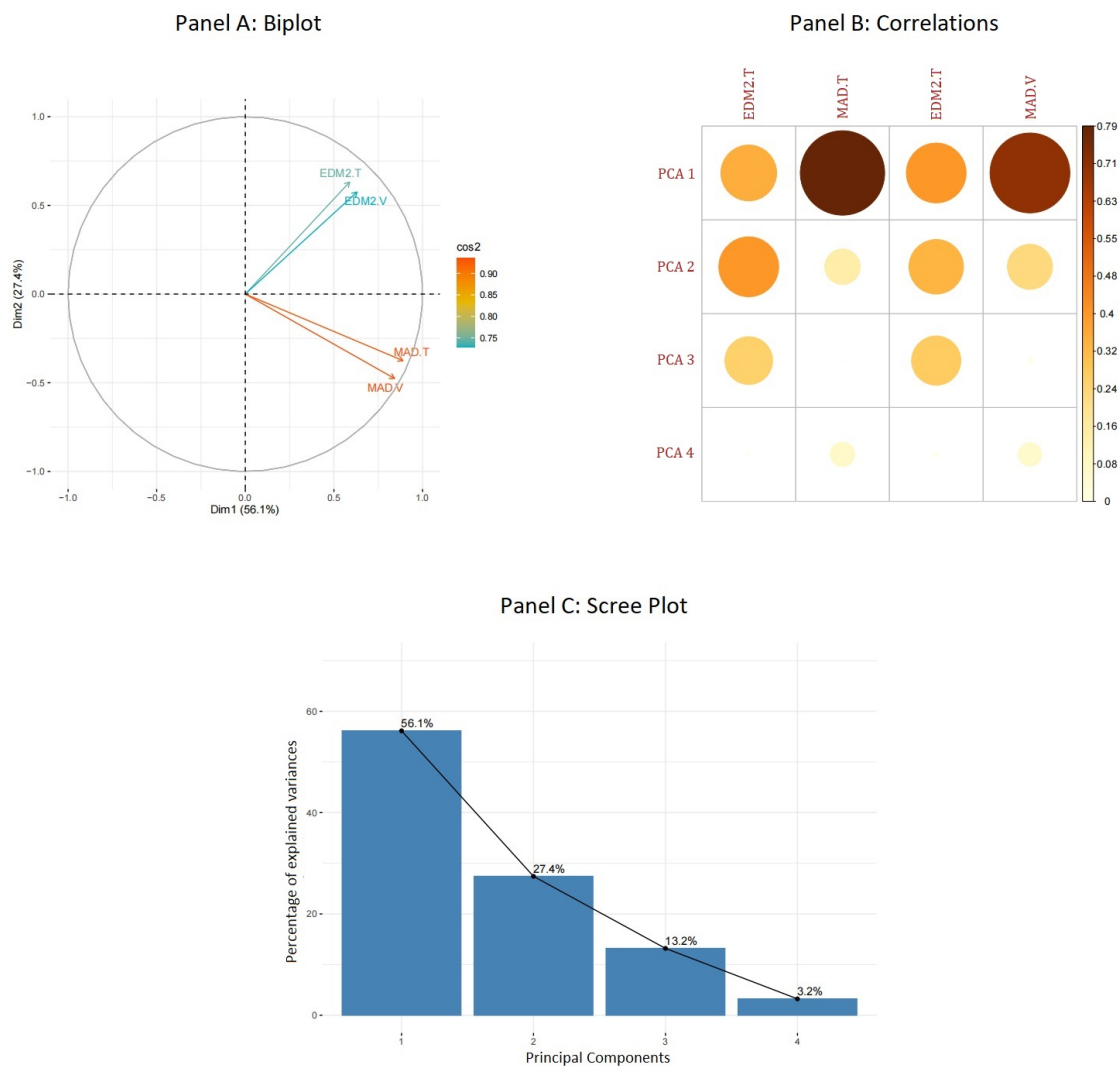


Figure A.4. Principal components results. The three panels show details on the derivation of the principal components measure. Panel A shows the biplot of the original fake measures, MAD and EDM. The extension .V stands for a measure based on volume series. The extension .T stands for a measure based on trades series. The vector direction and location show their correlations with the first two principal components of the combined measures. Panel B shows the correlations of the four original measures with the four dimensions of the derived principal components measure. The size of the circle indicates the contribution of the original measure to the dimension of the principal components. The color of the circle indicates the strength of the correlation. Panel C shows the scree plot of the four first dimensions of the principal components measure. The height of the bars indicates the size of the explained variation of that principal component dimension to the original measures of fake trading.



Appendix B: Variable definitions and additional empirical results

Table B.1. Variable description

Variable	Type	Description
<i>Market</i>		
CMC Listed: Coverage	Continuous (0, 1]	The monthly fraction of market capitalization of cryptocurrencies covered in our sample to the aggregate crypto market capitalization from www.CoinMarketCap.com
Currencies	Integer	Total number of cryptocurrencies (tokens and coins) listed on at least one exchange in a given month
Currencies: Entry	Integer	Total number of cryptocurrencies listed on at least one exchange in a given month and not listed on any exchange in the previous month
Currencies: Exit	Integer	Total number of cryptocurrencies not listed on any exchange in a given month and listed on at least one exchange in the previous month
Currency pairs	Integer	Total number of distinct currency pairs listed on at least one exchange in a given month
Currency pairs: Entry	Integer	Total number of currency pairs listed on at least one exchange in a given month and not listed on any exchange in the previous month
Currency pairs: Exit	Integer	Total number of currency pairs not listed on any exchange in a given month and listed on at least one exchange in the previous month
Volume: \$U.S. (MM)	Continuous (0, ∞)	Aggregate reported volume of trading in all currency pairs on all exchanges in a given month in billions of \$U.S.
# Trades (M)	Integer	Aggregate reported number of trades in all currency pairs on all exchanges in a given month in millions
Exchanges	Integer	Number of exchanges listing at least one currency pair in a given month
Exchanges: Entry	Integer	Number of exchanges listing at least one currency pair in a given month and not listing any currency pairs in the previous month
Exchanges: Exit	Integer	Number of exchanges not listing any currency pairs in a given month and listing at least one currency pair in the previous month
HHI Exchanges: # Currencies	Continuous (0, 1]	Sum of squared number of currencies listed on each exchange divided by the squared sum of number of currencies listed on all exchanges in a given month

Table B.1. Variable description – continued

Variable	Type	Description
<i>Exchanges</i>		
Age	Integer	The difference in months between current month and the first month any currency pair was listed on an exchange
Currency pairs	Integer	Number of currency pairs listed on an exchange in a given month
Currency pairs: Entry	Integer	Number of currency pairs that are listed on an exchange in a given month that were not listed on the exchange in the previous month
Currency pairs: Exit	Integer	Number of currency pairs that are not listed on an exchange in a given month that were listed on the exchange in the previous month
AML	Indicator	Equals one if an exchange implemented an AML policy and provides detailed information about conformity with accepted international AML procedures
KYC	Indicator	Equals one if there are evidence that the exchange provides clear guidelines, requires documents, and verifies sources of clients' funds
Crypto-friendly location	Indicator	Equals one if an exchange is located in Singapore, Russia, Estonia, Malta, Luxembourg or Switzerland
Traffic (MM)	Integer	Number of monthly visits (in billions) to an exchange's website, as reported by www.similarweb.com
Twitter retweets	Integer	Number of Twitter retweets of tweets by an exchange's official Twitter account in a given month
Github commits	Integer	The amount of code reviews (commits) at the exchange main Github repository in a given month
Africa, Asia (ex-China), China, Central and South America, North America, Eastern Europe, Western Europe (ex Islands), and Europe: Offshore	Binary	Indicator of headquarters in a country belonging to a respective region
<i>Currency pairs</i>		
Listed on # exchanges:	Integer	The number of exchanges on which a currency pair is listed in a given month
log (Listed on # exchanges):	Continuous $(0, \infty)$	Natural logarithm of the number of exchanges on which a currency pair is listed in a given month
Age of listing on any exchange	Integer	Difference in months between current month and the first month a currency pair was listed on any exchange
Age of listing on a given exchange	Integer	Difference in months between current month and the first month a currency pair was listed on a particular exchange
Token	Indicator	Equals one for tokens and zero for coins

Table B.1. Variable description – continued

Variable	Type	Description
<i>Quality measures</i>		
MAD: Volume	Continuous (0, 1)	Mean absolute distance between the frequency of leading digits of the trading volume series in a currency pair on a given exchange in a given month and the frequency of leading digits given by Benford's Law
MAD: # Trades	Continuous (0, 1)	Mean absolute distance between the frequency of leading digits of the number of trades series aggregated at the ten-minute frequency in a currency pair on a given exchange in a given month and the frequency of leading digits given by Benford's Law
EDM: Volume	Integer	The number of structural breaks in series of trading volume within ten-minutes intervals of a currency pair on a given exchange in a given month identified by E-Divisive with Medians algorithm
EDM: # Trades	Integer	The number of structural breaks in series of number of trades within ten-minutes intervals involving a currency pair on a given exchange in a given month identified by E-Divisive with Medians algorithm
<i>Principal components</i>		
PC1: Volume	Continuous	The first principle component of Mad: Volume, and EDM: Volume
PC1: # Trades	Continuous	The first principle component of Mad: # Trades, and EDM: # Trades
PC1: Both	Continuous	The first principle component of Mad: Volume, EDM: Volume, Mad: # Trades, and EDM: # Trades

Table B.2. Exchange location. Area of main operations and headquarter country are from www.CoinGecko.com and www.Cointelligence.com.

Exchange	Main operations region	Headquarter country
ACX	Asia	Australia
Allcoin	North America	Canada
AnyBits	Western Europe	Ireland
BeQuant	Western Europe	Malta
Bibox	Asia	China
BigONE	Africa	Seychelles
Binance	Western Europe	Malta
Bitbank	Asia	Japan
BitBay	Eastern Europe	Poland
Bitfinex	Western Europe	BVI
bitFlyer	Asia	Japan
BitForex	Africa	Seychelles
Bitibu	Asia	Turkey
Bitlish	Western Europe	UK
BitMEX	Africa	Seychelles
Bitso	Others America	Mexico
Bitstamp	Western Europe	UK
Bittrex	North America	USA
Bit-Z	Asia	Hong Kong
BTC-Alpha	Western Europe	UK
BTC-e	Eastern Europe	Russia
BtcTurk	Asia	Turkey
bx.in.th	Asia	Thailand
CEX.IO	Western Europe	UK
Cobinhood	Asia	Taiwan
Coinbase	North America	USA
CoinEx	Western Europe	UK
CoinMate	Eastern Europe	Slovakia
Ethfinex	Western Europe	BVI
EXX	Asia	China
FTX	Others America	Bahamas
Gatecoin	Asia	Hong Kong
Gemini	North America	USA
HitBTC	Asia	Hong Kong
Huobi	Asia	China
Kraken	North America	USA
KuCoin	Africa	Seychelles
LocalBitcoins	Western Europe	Finland
OkCoin	Asia	China
OKX	Asia	Belize
Poloniex	North America	USA
Quoine	Asia	Japan
SouthXchange	Others America	Argentina
TheRockTrading	Western Europe	Italy
TideBit	Asia	Hong Kong
Tidex	Eastern Europe	Russia
UPbit	Asia	South Korea
Yobit	Eastern Europe	Russia
Zaif	Asia	Japan
ZB	Asia	China

Table B.3. CMC Adjusted Volume: Exchanges average CMC (www.coinmarketcap.com) adjusted ranking drop between the three months before and three months after the CMC index implementation.

CMC Rank Change			
Exchange	Top Losers	Exchange	Top Winners
Bibox	1	Binance	1
Bittrex	2	Bitstamp	2
Kucoin	3	Coinbase	3
Bitfinex	4	Poloniex	4
Cex-io	5	Bitforex	5

Table B.4. Average measures of anomalous trading within three months prior to CoinMarketCap index introduction.

Rank	Exchange	PCA	MAD	EDM
Below-median PCA				
1	KuCoin	-0.21	0.12	0.09
2	ZB	-0.36	-0.07	0.03
3	Tidex	-0.45	-0.18	-0.33
4	BTC-Alpha	-0.50	-0.35	-0.28
5	HitBTC	-0.57	-0.48	-0.22
6	Binance	-0.57	-0.19	-0.29
7	Coinbase	-0.60	-0.33	-0.47
8	TheRockTrading	-0.67	-0.13	-0.51
9	Bitbank	-0.70	-0.27	-0.52
10	Poloniex	-0.75	-0.36	-0.50
11	Bittrex	-0.81	-0.46	-0.40
12	bitFlyer	-0.82	-0.47	-0.34
13	Bitstamp	-0.82	-0.39	-0.55
14	Kraken	-0.84	-0.44	-0.48
15	Bitso	-0.87	-0.61	-0.36
16	UPbit	-0.88	-0.54	-0.44
17	Yobit	-0.92	-0.63	-0.58
18	CEX.IO	-0.94	-0.50	-0.49
19	Quoine	-0.94	-0.55	-0.62
20	Bitfinex	-0.94	-0.54	-0.53
21	SouthXchange	-1.04	-0.65	-0.63
22	BtcTurk	-1.05	-0.70	-0.46
23	OkCoin	-1.05	-0.72	-0.55
24	BitBay	-1.06	-0.69	-0.59
25	Gemini	-1.07	-0.66	-0.57
26	CoinMate	-1.10	-0.70	-0.58
27	Zaif	-1.12	-0.70	-0.65
28	TideBit	-1.22	-0.74	-0.71
Above-median PCA				
1	Bit-Z	0.68	0.82	0.28
2	OKX	0.80	1.13	0.37
3	BitForex	0.82	0.42	0.95
4	BeQuant	1.01	-0.24	2.46
5	EXX	1.19	0.79	0.92
6	Huobi	1.31	2.02	-0.19
7	CoinEx	2.16	2.62	1.59
8	Bibox	2.33	2.14	2.94
9	BigONE	3.17	3.22	3.20

Response Letter

Dear Professor Capponi (the Editor) and the Associate Editor,

We would like to thank the editorial team and the referees again for another outstanding set of comments, which helped us further improve and streamline the paper's arguments, and guided us in very useful additional robustness analyses.

In the revision, we tried our very best to address every remaining comment of all three referees. Some of the changes involved heavy rewriting of significant parts of the paper; others involved changes and additions to the empirical analyses; yet others focused on making discussions and explanations clearer and more precise.

We think that this revision resulted in a much tighter, more focused, and overall more convincing paper. We hope that you and the referees agree.

In what follows, we respond in detail to all the points raised by the referees (the referees' points are italicized and our responses follow each point).

Response to Referee I

Thanks for the opportunity to review “Cooking the Order Books: Information Manipulation and Competition among Crypto Exchanges”. This version is largely improved in terms of data coverage, empirical analysis, and writing. The data period was expanded from 76 months to 97 months, updated to May 2021. More data sources are included. The introduction of the CoinMarketCap index is added as a new exogenous-shock-based analysis. More importantly, in the revised manuscript, a new concept, “information manipulation,” is introduced and related to anomalous trading in the context of crypto exchanges. I appreciate the great effort the authors have put into the revision. However, I have a few concerns that remain unsolved from last revision. My comments are as follows.

Major concern

In the previous round of review, multiple reviewers/editors suggested the authors to ditch the fake trading idea and focus on exploring the product quality problems in cryptocurrency industry. The authors changed ‘fake trading’ to ‘volume inflation’, ‘anomalous trade volume’ and argued they are different though some detailed definitions. I think the paper is still mainly working on the detection and effect of fake trading. This remains largely unchanged with previous version, which does not comply with the suggestions from multiple reviewers/editors.

The authors gave up on the product quality idea. They introduced a new one-information manipulation, and attempted to interpret fake trading using the information manipulation theory. Compared to the product quality idea in the previous version, I agree that information manipulation could potentially be more closely related to fake trading and have an important implication in related topics. However, I feel the idea was not executed seriously. Instead, it is just used as a decoration, like the product quality idea in previous version.

I only found strong presence of information manipulation in the introduction section. It is not carefully defined, measured and examined in the main text of the paper; neither was it argued in the context of crypto exchanges. And the connection between this paper with previous information manipulation related studies are vague.

I’m not satisfied about this ‘put old wine in new bottle’ type of revision in general.

Thank you for this comment. We agree that in the previous draft we have not articulated the new idea (which we don’t think is of the “put old wine in new bottle” type) in the best way possible. The idea consists of the following logical steps:

- There is evidence in existing literature that firms manipulate information;
- Theoretically and intuitively, firms should engage in more information manipulation when facing more intense competition;
- Empirically, it is not entirely clear whether this relation is observed.

- Thus, our first research question is whether information manipulation is related to competition empirically (also in the causal sense);
- Our second research question is whether information manipulation has positive or negative short-run and long-run effects on (operating) performance;
- In the context of crypto exchanges, a crucial piece of information about an exchange is trading volume on it, thus this is the type of information that is frequently manipulated on/by crypto exchanges by means of fake/wash trading;
- Fake trading is not directly observable, thus it needs to be estimated, which we do before setting to answer the aforementioned two research questions;
- Our estimation is based on 1) identifying instances of anomalous volume, 2) examining whether anomalous volume is associated with exchanges' incentives to fake trades (see more on that in our reply to your question 2 (a) below).

We now discuss the theoretical and empirical relation between information manipulation and competition in the introduction (see the second paragraph on p. 1, where we also discuss existing empirical and theoretical papers on information manipulation; and in the beginning of Section 6 (see p. 21).

Apart from the major concern listed above, there are some detailed suggestions/confusions listed below for the authors to consider.

1. Data Selection and Filtering

- (a) *The current version focuses on the “top 50 exchanges”. The selection criteria are ambiguous. I am not sure by which criteria exchanges are sorted. Reputation? Liquidity? Volume? Or just CoinMarketCap rank? This is important because the authors are talking about volume inflation in the paper so different criteria affect rank hugely.*

We now added to the *Data* section the following information regarding our sample selection criteria (see Section 2.1 on pp. 8-9).

The choice of exchange coverage by our main data provider, Kaiko, incorporates the following aspects:

Compliance: Soundness of exchange policies, such as KYC/AML, market surveillance, country risk, financial regulation, and insurance coverage.

Business: Maturity and quality of the exchange's business, measured by the exchange's age, the strength of the team, and the product offerings.

Technology: The capacity of the exchange's platform to provide market-level information through widely adopted distribution standards, while ensuring sufficient level of infrastructure reliability.

Data Quality: The breadth of public market data (trades, OHLCV (open, high, low, close, volume), and order books), and the capacity of the exchange to provide real-time feeds and/or historical data.

Security: The capacity of the exchange to provide sufficient level of security and protection to its users while mitigating the occurrence of security breaches.

Liquidity: The level and stability of trading activity but also the quality and reliability of the liquidity publicly reported by the exchange.

We apply the following filters to form the sample of exchanges covered by Kaiko: (1) Centralized exchanges, (2) Spot trading, and (3) Must contain at least one pair at any point in time during our sample period in which the base asset is one of the following: BTC or ETH or USDT. These filters result in inclusion of 50 exchanges into the sample.

- (b) *The authors stated, "The mean number of distinct cryptocurrencies traded on exchanges in a given month, covered in our data, is 944, whereas at the end of our sample period, 3,112 currencies were traded." Both numbers seem very large to me. According to Coin-MarketCap (January 2023), I only found several exchanges (among hundreds) trading more than 944 currencies. Does the sample in this study include many rarely traded currencies? Or do you include decentralized exchanges or derivative exchanges?*

In the previous draft, the discussion of the summary statistics was not sufficiently clear. 944 (3,112) refers to the mean (maximum) number of distinct cryptocurrencies traded *collectively* on centralized exchanges in our sample in a given month. So, for example, if assets A, B, and C are traded on exchange 1 during a given month, and assets C, D, and E are traded on exchange 2 during that month, then the total number of distinct cryptocurrencies traded on exchanges in that month is 5 (A, B, C, D, and E). We now provide a clear explanation of these values: "The mean number of distinct cryptocurrencies traded on all exchanges in our sample (collectively) in a given month is 944..." (see the first paragraph of Section 3 on p. 10).

2. Measurement of fake trading

- (a) *Compared to the previous version, the authors replace "fake trading/wash trading" with volume inflation/ anomalous trading volume. I think a simple name changing is not necessary. You can just say it is 'an alternative measurements for fake trading'. Unless the authors want to discuss in depth what is the difference between fake trading and nonfake volume inflation. The criticism from previous round of review were on the rationality and solidity of measurement methodologies and additional contribution on previous literature, not on the names. Also, there is a paradox the authors need to explain, if the anomalous volume is not fake, how it is related to information manipulation?*

Thank you for this comment. It highlights the fact that in the previous draft, we have not explained these differences fully. The source of the confusion lies in that we have been using various terms interchangeably, sometimes when this was not warranted. Let us now clarify the terms.

We view *fake/wash trading* and *volume inflation* as synonyms: volume is inflated by means of fake trading. *Anomalous volume* is different, as it refers to statistical properties of volume (or trades) series that are significantly different from normal patterns. Anomalous volume may or may not be a result of purposeful volume inflation through wash trading. In order to establish this link (i.e. claim that anomalous volume is likely a result of fake trading), we perform tests in Section 4.4, in which we examine cross-sectional and time-series relations between (changes in) measures of anomalous volume (*not* of fake trading) and (changes in) exchanges' *incentives* to inflate volume. Only after establishing this link, we can (in principle) use fake/wash trading, volume inflation, and anomalous volume interchangeably. However, to avoid further confusion, we separate these terms throughout the paper.

We explain this logic in the introduction (see the first full paragraph on p. 4), in the beginning of Section 5 (see p. 15), and summarize it at the end of Section 5.4 (see end of p. 20 and beginning of p. 21).

- (b) *The authors measure anomalous trading using Benford's Law (MAD), detecting structural breaks in trade pattern (EDM) and Principal components analysis-based measurement (PCA), which aggregates MAD and EDM. I understand that MAD and EDM measure the fake trading from different perspectives but not sure of the necessity of PCA measurement. Therefore, I suggest including the necessity and advantages of the PCA method in the 4.3 section, considering PCA is an essential measurement in the analysis.*

We added the following explanation to Section 4.3 (*Aggregating anomalous trading measures*) (see p. 14):

Utilizing PCA in our analysis has at least three key advantages over employing raw measures directly. First, PCA consolidates the majority of the original variance, proving beneficial when there is substantial disagreement between the two measures. Second, PCA effectively addresses the issue of multicollinearity among raw measures, which is particularly significant given that we derive individual measures from volume and number of trades series that are highly correlated. Third, PCA standardizes the data, making linear analysis more intuitive.

We elaborate on the suitability of PCA measure on pp. 45-46 and footnotes 28 and 29 in Appendix A. Despite relying on PCA as our primary indicator of fake trading, we present the raw measures—MAD and EDM—in all tables to ensure robustness.

3. Validation of measurement

In 4.4.3 and 4.4.4 [5.3 and 5.4 in the new draft], the authors actually employ the Difference-in-differences (DiD) approach in analyzing the effect of the Chinese ban and the CoinMarketCap index. The basic logic of DiD is based on two groups and two periods. In the first period, none of the groups is exposed to treatment (policy). In the second period, only one group is affected by the policy and referred to as the treatment group. The control group doesn't get exposed to treatment.

- (a) *In both the Chinese ban and CoinMarketCap index cases, it needs to be clarified when the events were introduced and how the treatment period is defined.*

Thank you for your comment. To help the reader navigate through these analyses, we have significantly expanded the discussion of the two exogenous shocks and their timelines in the Introduction (see pp. 5-6).

- (b) *"Our sample period for this analysis includes the first three quarters of 2017 (the "pre-ban period") and the first two quarters of 2018 (the "post-ban period")". Does the sample exclude the period when the Chinese ban/CoinMarketCap index was implemented, why is it? I think the Chinese ban/CoinMarketCap index may have a larger impact during the implementation period than the post-event period when everything is settled.*

Thank you for your suggestion. We changed the post-ban period for the Chinese ban analysis to include the last quarter of 2019 (i.e. there is now no gap between the pre-ban and post-ban periods) in both the Chinese ban and the CoinMarketCap index analyses (in Tables 4 and 5, respectively). The results are robust to the change in post-ban window and have not changed significantly.

- (c) *In 4.4.3 Chinese ban section, "the control group includes exchanges that were operating in Pacific Asia at the time of the ban". I am confused in this part as most crypto exchanges operate world-wide in my knowledge. I am also unclear why the exchanges operating outside Pacific Asia are not included in the control group.*

We have changed "were operating" to "were headquartered", according to www.CoinGecko.com and www.Cointelligence.com. We also list all the control exchanges in footnote 20 on p. 18. We have also expanded the discussion in the Introduction of our tests centered around exogenous shocks (see pp. 5-6).

Our decision to use exchanges headquartered in Pacific Asia is aimed at selecting a control group that is more likely to have social, political, and economic similarities with China. However, following your suggestion, we performed an additional robustness analysis, in

which the control group includes all exchanges that were headquartered outside of China at the time of the ban. The new Table 4 now includes two panels. In Panel A ("Outside China") all non-Chinese exchanges constitute the control group, whereas in Panel B ("Pacific Asia") the control group includes only exchanges headquartered in Pacific Asia. The main results (i.e. the coefficients on Treated $\times$ Post) are robust to the composition of the control sample. See discussion on in the last paragraph on p. 18 and the first paragraph on p. 19.

- (d) *For the CoinMarketCap Liquidity index, I think it would be better to give more details about it. For example, what the index reflects and what factors have been utilised in the construction of the index, and the limitations.*

We now explain in much greater detail (see footnote 21 on p. 19) the determinants of the CoinMarketCap index:

CoinMarketCap's Liquidity Score is designed to capture expected slippage (or price impact) of trades. CoinMarketCap's methodology for constructing the index relies on estimating expected price impact of various trades. CoinMarketCap simulates trades drawn from the range typical of trade sizes by retail traders on centralized exchanges—between \$100 to \$200,000; simulates the price impact of market orders of these trades; and aggregates the results across various order sizes and currency pairs on a particular exchange. The smaller the resulting weighted average price impact the higher the liquidity score.

- (e) *In the DID analysis, there are other closely related factors that are worth considering during the introduction of the index, which might pollute the results. For example, CoinMarketCap was acquired by Binance in early 2020, around the time when the index was developed and introduced. Would Binance put some pressure on this index to push competitors? Around the launching of the index, CoinMarketCap also forced crypto exchanges to provide API for data collection, and implementing other measures on its ranking system including web visitors, trust scores etc. which may affect fake trades too.*

While it is impossible to rule out confounding effects of CoinMarketCap acquisition by Binance around the CoinMarketCap index introduction, note that our identification strategy relies not on the introduction of the index per se, but on differential effects of the index introduction on exchanges with relatively high/low levels of estimated anomalous trading volume prior to the index introduction. The idea behind this test is that the introduction of the index diminishes the importance of trading volume in determination of exchange rankings, lowering exchanges' incentives to manipulate information and inflate volume. This is true regardless of whether the index itself is unbiased. We now discuss this point on on p. 19 and footnote 22.

- (f) *In the 4.4.4 CoinMarketCap index section, “We define treated exchanges as those with top-ten anomalous volume estimates in the three months prior to December 2019 and control exchanges as those with bottom-ten anomalous volume estimates during the same period”. The index provides a new rank for all exchanges. All crypto exchanges listed in the CoinMarketCap would be affected by the introduction of the new rank. It is strange to classify control and treatment group using the authors criteria. It makes more sense to compare the exchanges listed on CoinMarketCap and those are not listed.*

As of February 18, 2014, 231 spot centralized exchanges are covered by CoinMarketCap. This number was even higher (over 500 exchanges) in 2020-2021. All of the 50 exchanges in our sample were covered by CoinMarketCap both before and after the index Introduction. Thus, it is impossible, unfortunately, to classify control and treatment groups based on CoinMarketCap coverage.

We performed two changes to this analysis. First, we added an analysis that relies on the differential effect of CoinMarketCap index introduction on the rankings of exchanges. In particular, we define treated exchanges as those that dropped most in the rankings as a result of the index introduction, whereas control exchanges are those whose ranking benefited the most from the index introduction. The idea is that exchanges that are penalized the most by the analysis of volume inflation would have reduced incentives to manipulate volume relative to exchanges that were penalized the least. See Panel A of Table 5 and the discussion on p. 20.

In addition, following a suggestion by another referee, we define control and treatment groups based on whether an exchange belongs to top/bottom five (as opposed to ten) in terms of abnormal volume pre-index-introduction and obtain results that are stronger than those based on original classification. See Panel B of Table 5 and discussion on p. 20.

- (g) *I am also interested in the relationship between the CoinMarketCap index and anomalous volume estimation. For example, does the exchange with high anomalous volume has a lower liquidity index?*

This suggestion is certainly interesting. Unfortunately, CMC does not maintain a historical record of crypto exchanges’ liquidity index ranks. Therefore, we are unable to conduct the suggested analysis.

4. *Competition and fake trading*

- (a) *Sections 5.1 and 5.2 [Sections 6.1 and 6.2 in the new draft] examine how competition among exchanges influences the extent of volume inflation. The indicators for competition and fake trades are contemporaneous, which makes it difficult for cause-and-effect reasoning. I suggest the authors use lagged value for the extent of competition instead and*

examine whether the competition among exchanges “motivates” the exchanges to fake trades.

We appreciate your suggestion. We have totally re-designed the tables in this section. We followed your suggestion and replaced the contemporaneous measure of static competition with the lagged one in Tables 6 and 7. Indeed, anomalous volume is positively related to lagged competition measure. The coefficients are larger and more significant than those on the contemporaneous measure used previously.

- (b) *I am not sure how information manipulation is related to competition and fake trades. The relation is not carefully explained in this section.*

Our analysis is based on the following building blocks. First, we hypothesize in the introduction (see p. 1) that optimal information manipulation should be increasing in competition intensity. Second, we show in Section 5 that our statistical measures of anomalous volume are likely to be correlated with true, unobservable fake trading (see conclusion in the last paragraph of Section 5 on pp. 20-21), which is a form of information manipulation by exchanges. Third, in Section 6, we set to examine the relation between information manipulation (proxied by estimated volume inflation) and the intensity of competition (see the beginning of Section 6 on p. 21, which precedes Hypothesis 1.)

5. Consequences of fake trading

- (a) *In section 6 [Section 7 in the new draft], the authors gauge the short-term and long-term effects of fake trades using web popularity and social media popularity. I think it makes some sense but is limited. If the authors are interested in digging more into the implication of fake trades, I suggest the authors evaluate the impact on other perspectives, such as market share, and revenue.*

We have struggled a lot with the question of which measures of performance that are 1) relevant and likely to be related to financial performance and 2) unbiased. Measures related to web and social media popularity are to a large extent unbiased. However, they are obviously indirect.

As you correctly suggest, More direct measures, are based on market share or revenues. However, there are reasons why these measures may be severely biased, reducing their usefulness. A reason against using a proxy for revenues, which is essentially the product of the exchange’s fee and its trading volume, is that such measure is increasing in the extent of volume inflation by the exchange. In earlier versions of this study, we have included this analysis but ultimately decided to shift our focus to web traffic and social

media as proxies for demand, as these measures are less likely to be targeted for manipulation. Using market share (based on trading volume or number of trades) is subject to the same concern of potentially inflated trading volume/number of trades, where the degree of volume inflation is different across exchanges. Thus, we decided to keep the analysis using web and social media popularity-based measures, while explaining this choice in the text (see discussion following Hypothesis 2 on pp. 24-25).

- (b) *In Tables 9 and 10 [Tables 8 and 9 in the new draft], I suggest include the lagged value of popularity in the regression as a control variable. Popularity begets popularity, which may be less affected by fake trades in one period.*

Thank you for this suggestion. We added lagged web popularity in Table 8 and lagged social media popularity in Table 9 (old Tables 9 and 10). Consistent with your conjecture, there is a positive autocorrelation in measures of popularity. Importantly, the qualitative results (i.e. the effects of fake trading on web popularity and social media popularity over various horizons) are robust to the inclusion of lagged web popularity, as can be seen in Tables 8 and 9, respectively.

Response to Referee II

Summary

The authors have done a great job revising the paper and addressing my concerns from the last round. The motivation and contribution of the new draft is much clearer. I especially like the refocusing of the paper to information manipulation and competition and appreciate the authors' efforts in extending the sample. I commend the authors.

We appreciate your kind words. In this revision, we did our best to address your remaining concerns, as well as those of other referees, the Associate Editor, and the Editor.

Comments

1. *In the last round, I asked about the results on fake trading and trading volume. In the new draft, the authors deleted the results in this regard. The authors argue (in the reply letter) that fake trading does not increase liquidity, but rather "1) it raises the realized price impact relative to an anticipated one, and 2) it increases uncertainty regarding price impact". I would suggest that the authors discuss more about this in the paper.*

Thank you for this suggestion. We now provide an extensive discussion in the introduction why fake trading does not increase real liquidity that can be used by traders: "Trading volume on crypto exchanges can be artificially inflated in two ways. First, an exchange can execute back-and-forth transactions among its own wallets by making an order and immediately taking it. Such orders appear in the order book for a split second but cannot be taken by independent, non-exchange-affiliated traders. Second, an exchange can create perverse incentives for traders and/or project owners to trade with themselves, e.g., by eliminating trading commissions for "top-tier" traders or by requiring projects to facilitate a certain minimal trading volume in order to maintain exchange listing, which the projects typically do with the help of market makers." We then argue that the real effects of fake trading are on the realized-versus-expected price impact and on trading cost uncertainty (see second paragraph on p. 2).

2. *Footnote 10 on page 35: "The list of crypto friendly countries includes Singapore, Russia, Estonia, Malta, Luxembourg and Switzerland." Although the authors provide a definition of crypto-friendly countries here, I wonder whether there is any more "official" definition of crypto-friendly countries and hope the authors can discuss more about this classification. The authors define "wide acceptance of cryptocurrencies" as crypto-friendly. Which type of cryptocurrencies you are referring to? Some countries, (China for example), allow digital currency (CBDC), but not really crypto-friendly. Also, just curious, it is said Australia is also crypto-friendly, but it is not in the list.*

Regrettably, there are no official definitions for crypto-friendly countries. Instead, a widely accepted definition among practitioners exists, which we rely upon when classifying countries (see <https://getgoldenvisa.com/crypto-friendly-countries> for instance). In addition, crypto friendliness changes over time, and we are not aware of data sources tracking its time series evolution. Also, even though crypto exchanges are headquartered in specific locations, they often employ intricate business structures that make it challenging to accurately determine their legal jurisdiction, especially since many exchanges operate internationally. For these reasons, we ultimately chose not to include crypto-friendly indicators in our analysis.

As an alternative to using crypto-friendly indicators, we opted for legal indicators (i.e. AML) and compliance indicators (i.e. KYC). We obtain exchange-level characteristics, including the adoption of anti-money-laundering measures (AML), the existence of know-your-customer requirement (KYC), from www.CoinGecko.com and www.Cointelligence.com. These indicators are more likely to effectively capture the extent of crypto exchange adherence to accepted international AML rules and best business practices (e.g., whether the exchange offers transparent guidelines, mandates documentation, and verifies the sources of clients' funds).

3. *This paper mainly uses Github commits (amount of code reviews) to denote technological development of exchanges. One concern here is that Github commits may only capture one dimension of technological development. Deng et al. (2018), have constructed a development composite index, which is calculated as the sum of several dimensions of development progress scores including Subscribers, Contributors, Proportion of Closed Issues, Pull Request Merged and so on. There are also some other papers that use different measures to capture other aspects of technological development. The authors should cite these papers and think more about (or at least provide more discussions on) how to fully capture multiple dimensions of technological development.*

We appreciate your suggestion. We begin by noting that the commits-based proxy we employ is quite highly correlated with the composite index from Deng et al. (2018), as documented in Lyandres et al. (2022).

Importantly, there is not much data available on the components of the composite index of Deng et al. (2018), i.e. the number of contributors (outside of the core team), the number of closed issues, and the number of pull requests. See, for example, footnote 9 in Lyandres et al. (2022): "In unreported analysis, we examine additional proxies for code development activity – the number of contributors (outside of the venture), the number of closed issues, the number of pull requests – as suggested by Deng et al. (2018). While these are reasonable alternative measures of the code revision activity, they are quite sparsely populated in our data: only 10% (16%, 14%) of observations with non-zero commits have non-zero contributors (closed issues, pull requests). The results using these alternative measures of code production are generally statistically insignificant, likely due to insufficient statistical power." We face a similar problem in our setting, and, thus, opted for our original measure based on commits.

Nonetheless, we acknowledge [Deng et al. \(2018\)](#) in footnote 12 on p. 11, where we posit that an analysis of all aspects of a crypto exchange's technological development could benefit from employing the composite index.

4. *Typo: Line 44 on page 36: deviations from Benford's Law, not "deviations fro Benford's Law"*

We fixed this typo and made every effort to ensure that there is no other typos across the manuscript.

5. *On page 43, "We define treated exchanges as those with top-ten anomalous volume estimates in the three months prior to December 2019 and control exchanges as those with bottom-ten anomalous volume estimates during the same period." The definition of treated group is a bit arbitrary here. It would be better if the authors can use a different definition for the robustness check, for example, top-five or top-twenty exchanges as treated group.*

We have implemented your suggestion. We now report results obtained while designating crypto exchanges with the top (bottom) 5 anomalous trading volume as the treatment (control) group. The effects are actually more accentuated for the top (bottom) 5 than for the top (bottom) 10 exchanges. The results are now reported in the re-designed Table 5, Panel B.

In addition, we now rely on the differential effect of CoinMarketCap index introduction on the rankings of exchanges. In particular, we define treated exchanges as those that dropped most in the rankings as a result of the index introduction, whereas control exchanges are those whose ranking benefited the most from the index introduction. The idea is that exchanges that were penalized the most by the estimation of volume inflation by CoinMarketCap would have lower incentives to manipulate volume relative to exchanges that were penalized the least. See the discussion on pp. 19-20 and the results in Panel A of Table 5.

6. *Typo: Line 23 on page 45: a pair is traded or "a pair is trading"?*

Thank you. We ensured consistent terminology ("pair is traded") throughout the paper. We have also made our best effort to make the writing consistent throughout.

Response to Referee III

This paper is a significant improvement over the previous version entitled “Competition and Product Quality: Fake Trading on Crypto Exchanges.” Specifically, I think the new version has moved away from its originally claimed contributions to product qualities and become more focused on volume manipulations on cryptocurrency exchanges. The authors have also addressed most of my other comments. Overall, I am happy to see the paper to have improved significantly.

As a remaining comment, I think the new title is still not accurate. Overall, volume manipulation alters trade records rather than order book (which is mostly about quotes). What about just simply replacing “cooking the order books” with e.g., “faking trades” or something more accurate?

We agree with you that the title in the previous draft was catchy but imprecise. The new title, which reflects the essence of the analysis of the paper is "Trading volume manipulation and competition among centralized crypto exchanges".

Finally, I would not insist on the following point, but I am just curious:

The authors explain that their Benford’s law measures differs from those in Cong et al in that theirs use the sum of absolute difference while the latter use Chi-square differences. In other words, this paper uses an L1 measure while Cong et al use an L2 measure. The authors also claim that their L1 measure may outperform the L2 measure by quoting Kossovsky (2021) and by resorting to the fact that the L1 measure has been used in other studies that involve panel data (like this one). Rather than resorting to the literature, I think a more direct response to my comment is simply produce both L1 and L2 results and make a direct comparison (maybe in the appendix) and let the results speak for themselves. Direct results, coupled with findings from the literature, would likely make a more convincing argument (and also additionally contributes to the literature discussed).

Thank you for your comment. We conducted tests comparing L1 and L2 measures, as shown in the table below. The test results indicate that L1 and L2 produce similar outcomes, in a difference-in-differences design, albeit with divergences only in the statistical significance in the coefficients of the variables of interest and the model’s R-square. The table below provides an example of comparisons of our (L1) measure with the L2 measure in Cong et al. (2022) within the context of tests involving the Chinese ban. L1 performs somewhat better in terms of fitting the data in this empirical model than L2. Similar results are obtained for the other tests.

Although interesting, we have chosen not to include a detailed discussion of this and similar tables in the paper for two main reasons. First, we believe that such a discussion would be more suitable in a dedicated paper focused on evaluating the most appropriate econometric approach, as demonstrated in Kossovsky (2021). Second, our paper heavily relies and builds on the work of Cong et al. (2022), which is referenced throughout the manuscript. We have no reasons or intentions to undermine in any way the insights and conclusions presented in Cong et al. (2022). We hope you agree with our

The Chinese ban results using Cong et al. (2022) Benford-Law-based measure of anomalous volume. The regressions are estimated at the exchange-pair-month level in similar fashion as Table 4. L1 reports regression estimates for the mean absolute differences between theoretical Benford and observed volume series — the metric used in the paper. L2 reports regression results for chi-square differences between theoretical Benford and observed volume series — the metric use in Cong et al. (2022). Standard errors (reported in parentheses) are heteroskedasticity consistent and clustered at the currency pair level. * Significant at 10 percent; ** Significant at 5 percent; *** Significant at 1 percent.

	Panel A: Outside China		Panel B: Pacific Asia	
	L1	L2	L1	L2
Indicators of Interest:				
Treated	0.86 *** (0.02)	0.16 *** (0.03)	0.73 *** (0.03)	0.05 (0.04)
Post	0.24 *** (0.01)	0.39 *** (0.02)	0.31 *** (0.03)	0.47 *** (0.04)
Treated × Post	-0.19 *** (0.03)	-0.09 * (0.05)	-0.40 *** (0.06)	-0.36 *** (0.08)
Exchange Controls	yes	yes	yes	yes
Currency Pair Controls	yes	yes	yes	yes
Quote FE	yes	yes	yes	yes
Obs.	26,178	26,178	11,698	11,698
Adj. r^2	0.41	0.21	0.14	0.08

reasoning regarding this matter. We now state in footnote 15 on page 13 that our results are robust using the distance measure based on Chi square differences.